Algorithms & Probability Recap

A complete recap of the contents taught in the Algorithms and Probability lectures at ETH Zürich, which are part of the Basisjahr's second semester.

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color code: title, weeks, theorem, important stuff

by Erik Girdauskas

AuW semester recap

Weeks 1-2:

k-connectivity: (considering a graph $G=(E,V)$)

→ u and v are **connected** if $\exists$ u-v-path in G

→ G is **connected** if for all $u,v \in V$ with $u \neq v$ $\exists$ u-v-path in G

→ u and v are **k-connected** if $|V| \geq k+1$ and $\forall X \in V \setminus \{u,v\}$ with $|X|<k$ is u-v-connected in $G[V \setminus X]$

→ G is **k-connected** if $|V| \geq k+1$ and for all $X \subseteq V$ with $|x|<k$, $G[V \setminus x]$ is connected

→ X is an **u-v separator** if u and v are in separate connected components of $G[V \setminus X]$

→ G is also **k-connected** if $|V| \geq k+1$ and for all $u,v \in V$ every u-v sep. X has $|X| \geq k$.

→ you have to delete at least k vertices to break connectivity

→ If G contains a vertex with degree $<k$, G is **not** k-connected

→ **Menger's Theorem (vertices):** every X has $|X| \geq k \iff \exists k$ **vertex disjoint** u-v paths

↑ I will generally write X for u-v separators

→ **Menger's Theorem (all vertices):** G is **k-conn.** $\iff$ for all $u,v \in V$ with $u \neq v$, $\exists k$ internally **vertex disjoint** u-v paths

→ X is a **u-v edge separator** $\Rightarrow$ u and v are in different connected components of $G' = (V, E \setminus X)$

→ G is **k-edge-connected** $\Rightarrow$ for all $u,v \in V$, all X have $|X| \geq k$

→ if all $X \subseteq E$ have $|X|<k$, $(V, E \setminus X)$ is connected (assuming G is k-edge-connected)

→ **Menger's Theorem (Edges):** G is **k-edge-connected** $\iff$ for all $u,v \in V$ with $u \neq v$, $\exists$ k **edge-disjoint** u-v paths

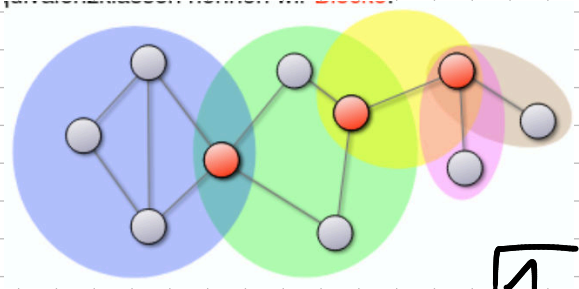
Cut vertices, cut edges: → $v \in V$ is cut vertex if (and only if) $G[V \setminus \{v\}]$ not conn., $e \in E$ is cut edge if (and only if) $G-e$ not conn.

→ $\{x,y\} \in E$ is a cut edge if $\deg(x)=1$ or $\deg(y)=1$ or x or y are cut vertices

Blocks: equiv. relation: $e \sim f \iff \begin{cases} e=f \text{ or} \\ \exists \text{ cycle with } e,f \end{cases}$

→ we define the **block graph** as $T=(A \cup \mathcal{B}, E_T)$ with $A = \{\text{cut vertices of } G\}$ and $\mathcal{B} = \{\text{blocks of } G\}$

→ $\forall a \in A, \forall b \in \mathcal{B}: \{a,b\} \in E_T \iff a$ adjacent to an edge in b



Theorem: if G is connected, T is a tree.

Finding cut vertices using DFS:

We use the following variation from the script:

DFS-Visit(G, v)

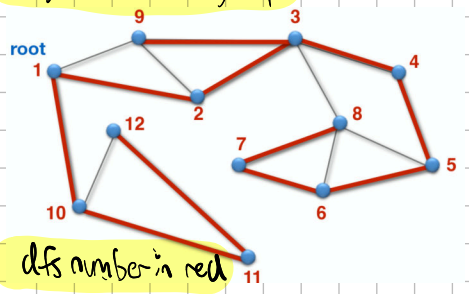
```
1:  num ← num + 1
2:  dfs[v] ← num
3:  low[v] ← dfs[v]
4:  isArtVert[v] ← FALSE
5:  for all {v, w} ∈ E do
6:      if dfs[w] = 0 then
7:          T ← T + {v, w}
8:          val ← DFS-Visit(G, w)
9:          if val ≥ dfs[v] then
10:             isArtVert[v] ← TRUE
11:         low[v] ← min{low[v], val}
12:     else dfs[w] ≠ 0 and {v, w} ∉ T
13:         low[v] ← min{low[v], dfs[w]}
14: return low[v]
```

DFS(G, s)

```
1: ∀v ∈ V :  dfs[v] ← 0
2: num ← 0
3: T ← ∅
4: DFS-Visit(G, s)
5: if s hat in T Grad mindestens zwei then
6:     isArtVert[s] ← TRUE
7: else
8:     isArtVert[s] ← FALSE
```

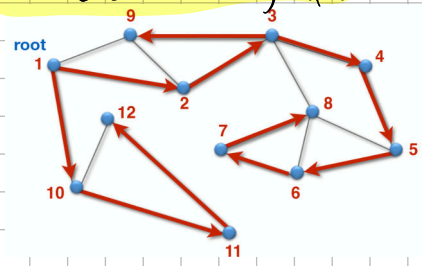
this works as follows:

regular DFS graph:

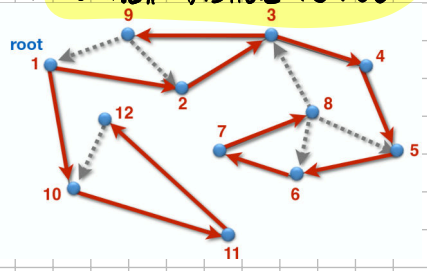


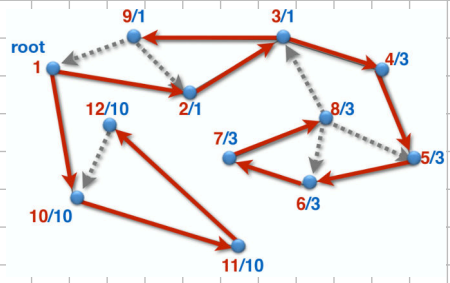
dfs number in red

directed DFS graph:



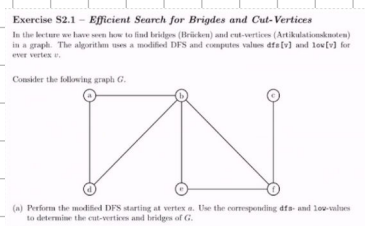
direct leftover edges away from their visited vertices





skills needed to do this on paper:

apply DFS on graph, write down DFS number (when we visit the nth vertex, n is the DFS number)
→ direct the edges and leftovers
→ start out with low[v] = dfs[v]
→ update for some leftover edges and when the algorithm backtracks


Exercise S2.1 – *Efficient Search for Brigdes and Cut-Vertices*

In the lecture we have seen how to find bridges (Brücken) and cut-vertices (Artikulationsknoten) in a graph. The algorithm uses a modified DFS and computes values dfs[v] and low[v] for ever vertex v.

Consider the following graph G.

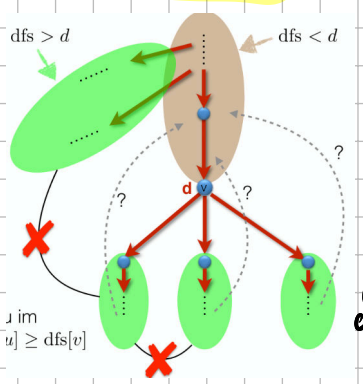
(a) Perform the modified DFS starting at vertex a. Use the corresponding dfs- and low-values to determine the cut-vertices and bridges of G.

We now define low[v] := smallest DFS number reachable through a directed path using an arbitrary amount of tree edges and max. 1 leftover edge

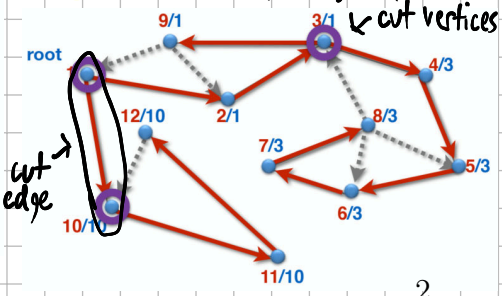
more formally: low values can be computed in $O(|V|+|E|)$

$$low[v] = \min\left(dfs[v], \min\begin{cases} dfs[w], \text{if } (v,w) \text{ leftover edge} \\ low[w] \text{ if } (v,w) \text{ tree edge} \end{cases}\right)$$

→ v cut vertex <=> 1) v ≠ root and v has child w in DFS tree with low[w] ≥ dfs[v]
2) v = root and v has at least 2 children in the DFS tree



in the example graph:



cut edges: a tree edge e = (v, w) (v parent, w child) is a cut edge if and only if low[w] > dfs[v].

Theorem: This modified DFS computes cut vertices and cut edges in a connected graph in O(n)

Cycles:

→ **Hamilton cycle** → cycle in G that contains every vertex exactly once

→ **Euler tour** → a closed walk in G that contains every edge exactly once

→ **Theorem:** a connected graph contains an Euler tour if and only if every vertex has an even degree and it can be found in $O(|E|)$

→ the lecture details a "runner and tortoise" algorithm where the runner first finds a closed walk, then the tortoise finds an edge-disjoint closed walk, and so on, until there are no edges (success, Euler tour) or there are no closed walks (no Euler tour)

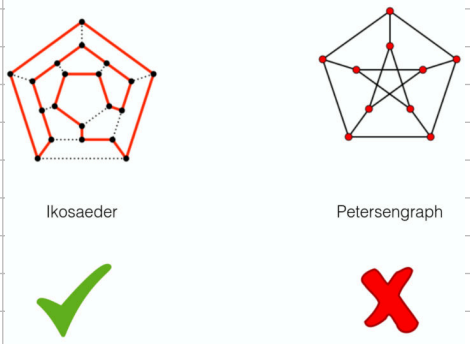
→ **algo. from script:**

```
EulerTour(G, v_start)
1: // Schneller Läufer
2: W ← RandomTour_G(v_start)
3: // Langsamer Läufer
4: v^langsam ← Startknoten von W.
5: while v^langsam ist nicht letzter Knoten in W do
6:     v ← Nachfolger von v^langsam in W
7:     if N_G(v) ≠ ∅ then
8:         W' ← RandomTour_G(v)
9:         // Ergänze W = W_1 + ⟨v⟩ + W_2 um die
10:        // Tour W' = ⟨v'_1 = v, v'_2, ..., v'_{ℓ-1}, v'_ℓ = v⟩
11:        W ← W_1 + W' + W_2
12:    v^langsam ← Nachfolger von v^langsam in W
13: return W
```

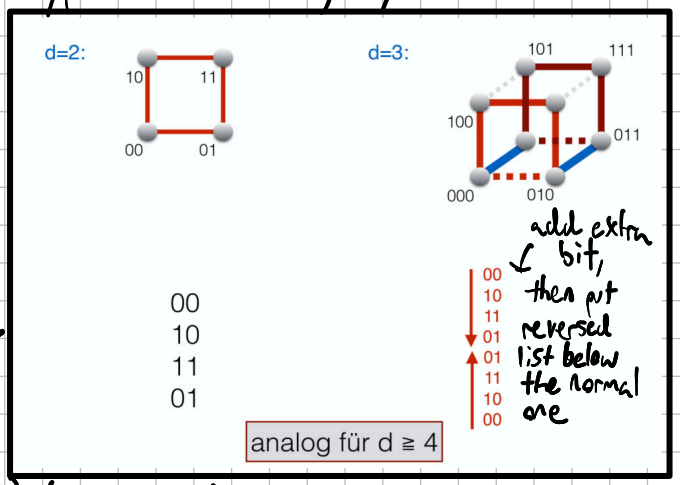
```
RandomTour_G(v_start)
1: v ← v_start
2: W ← ⟨v⟩
3: while N_G(v) ≠ ∅ do
4:     Wähle v_next aus N_G(v) beliebig.
5:     Hänge v_next an die Tour W an.
6:     e ← {v, v_next}
7:     Lösche e aus G.
8:     v ← v_next
9: return W
```

Hamilton cycles:

use these as **counterexamples** in proofs: **Theorem:** an $n \times m$ grid contains a Hamilton cycle **if and only if** $n*m$ is even.



Hypercubes and gray codes:



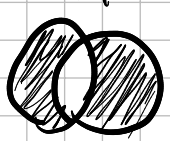
Theorem (Dirac): If $|V| \geq 3$ and min. degree $\delta(G) \geq |V|/2$, G contains a Hamilton cycle.

Sieve formula (Inclusion/Exclusion): for finite sets $A_1, \dots, A_n$ $(n \geq 2)$:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{l=1}^{n} (-1)^{l+1} \sum_{1 \leq i_1 < \dots < i_l \leq n} |A_{i_1} \cap \dots \cap A_{i_l}| = \sum_{i=1}^{n} |A_i| - \sum_{1 \leq i_1 < i_2 \leq n} |A_{i_1} \cap A_{i_2}|$$

$$+ \sum_{1 \leq i_1 < i_2 < i_3 \leq n} |A_{i_1} \cap A_{i_2} \cap A_{i_3}| - \dots + (-1)^{n+1} \cdot |A_1 \cap \dots \cap A_n|$$

for $n=2$: $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$



Theorem: One can decide and, if applicable, find a Hamilton cycle in $O(|V|^2 \cdot 2^{|V|})$ (if a Hamilton cycle ex.)

Theorem (Karp): Deciding if a graph contains a Hamilton cycle is NP-complete

Hamilton algorithm (DP): Time complexity: $n^2 \cdot 2^n$, Space: $n \cdot 2^n$

Idea: For all $S \subseteq [n]$ with $1 \in S$ and all $x \in S$ with $x \neq 1$:

$P_{S,x} := \begin{cases} 1, & \exists\ 1\text{-}x\text{-path containing exactly the vertices from } S \\ 0 & \text{else} \end{cases}$

Initialization: $\forall x \in \{2,\dots,n\}$: for $S = \{1,x\}$ set $P_{S,x} = 1$ if and only if $\{1,x\} \in E$.

Loop: for all $s = 3$ to n
for all $S \subseteq [n]$ with $1 \in S$ and $|S| = s$:
for all $x \in S$ with $x \neq 1$:

$$P_{S,x} = \max\{P_{S\setminus\{x\},y} : y \in N(x) \cap S,\ y = 1\}$$

Output: G contains Hamilton cycle $\Leftrightarrow \exists\ x \in N(1)$ with $P_{[n],x} = 1$

Traveling Salesman: given a complete graph with n vertices and the distance $\ell : \binom{[n]}{2} \to \mathbb{R}$ between any 2 vertices, find the smallest round trip:

$\min_{H \text{ Hamilton cycle}} \sum_{e \in E(H)} \ell(e)$ → find a round trip smaller than a given integer X.

It is NP-complete: Graph $G = ([n], E) \Leftrightarrow$ complete graph $K_n = ([n], \binom{[n]}{2})$
with length function $\ell : \binom{[n]}{2} \to \{0,1\}$
s.t. $\ell(\{x,y\}) = \begin{cases} 0, & \text{if } \{x,y\} \in E \\ 1 & \text{else} \end{cases}$

Then: G includes Hamilton cycle $\Leftrightarrow opt(K_n, \ell) = 0$

From this: there cannot be a polynomial C-approx. algo. for any $C > 0$

we can derive the **metric TSP** if:

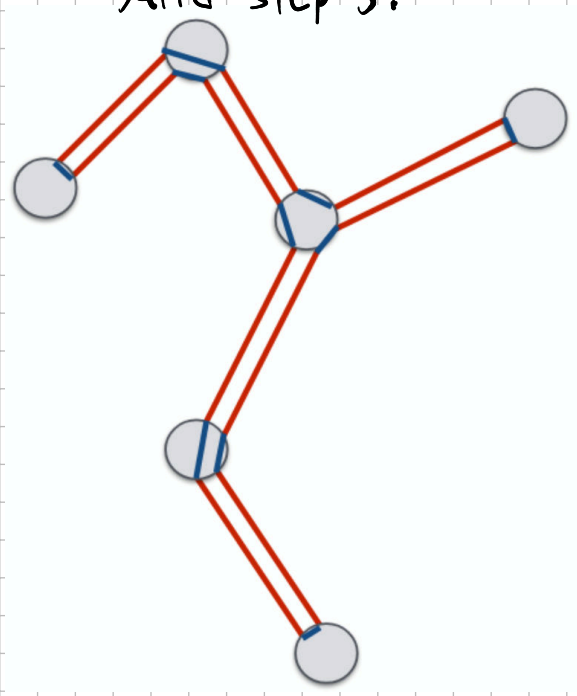
ℓ fulfills **triangle inequality**: $\ell(x,z) \leq \ell(x,z) + \ell(y,z)\ \ \forall x,y,z \in [n]$

→ we can derive a **2-approximation** using the MST

Algo.: 1. find MST T $(\ell(T) \leq opt(K_n, \ell))$
2. double all edges of T $(2\ell(T) \leq 2opt(K_n, \ell))$
3. find Euler tour W $(\ell(W) = 2\ell(T) \leq 2opt(K_n, \ell))$
4. go through W, use any shortcuts s.t. any vertex is visited only once

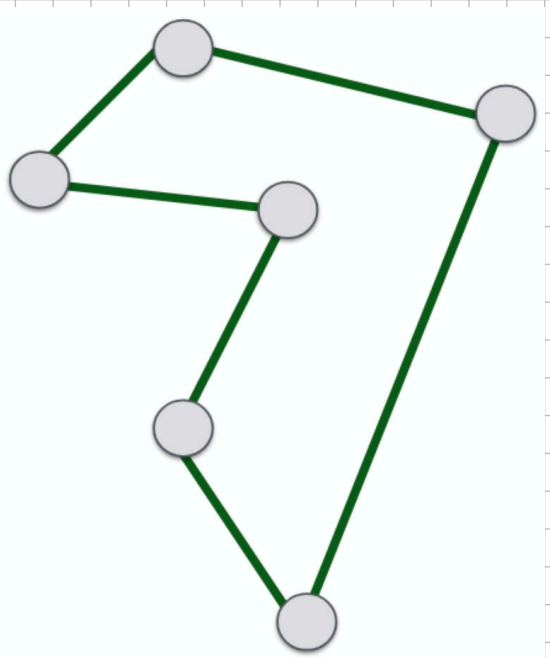
→ Hamilton cycle C, $\ell(C) \leq \ell(W) = 2\ell(T) \leq 2opt(K_n, \ell)$

After step 3:



Step 4 →

this is allowed because we have a complete graph



Weeks 3 and 4:

Matchings: → $M \subseteq E$ is a matching if no vertex is adjacent to more than one edge of M.

→ M covers v if $\exists e \in M$ that contains v

→ **perfect matching:** $|M| = |V|/2$ (every vertex is covered)

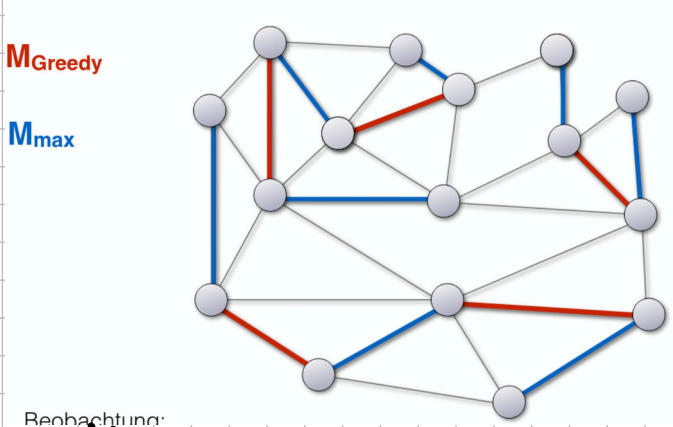
→ **maximal matching:** $\nexists M'$ with $M \subseteq M'$ and $|M'| > |M|$

→ **maximum matching:** $\exists M'$ with $|M'| > |M|$ (stronger condition)

→ **Theorem:** with the greedy algorithm, we can find a maximal matching M_{Greedy} with $|M_{Greedy}| \geq |M_{max}|/2$ (M_{max} is maximum)

GREEDY-MATCHING (G)

1: $M \leftarrow \emptyset$
2: **while** $E \neq \emptyset$ **do**
3: wähle eine beliebige Kante $e \in E$
4: $M \leftarrow M \cup \{e\}$
5: lösche e und alle inzidenten Kanten in G



Observations: →

→ for any edge in M_{max}: at least one of two vertices adjacent is covered by an edge from M_{Greedy}

→ any edge from M_{Greedy} can at most cover 2 edges from M_{max}

Augmenting paths: an M-augmenting path alternates between edges in M and not in M and starts/ends with edges not in M.

→ we can increase M's size by swapping along $M \Rightarrow M' := M \oplus P$ (XOR)

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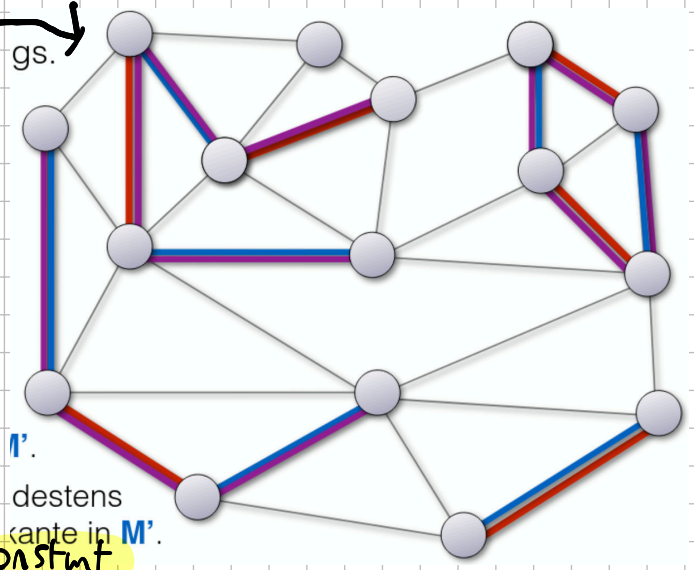
What happens if you have $M \oplus M'$ with M and M' being arbitrary matchings?

Observations:

- every vertex has degree ≤ 2
  $\Rightarrow$ collection of paths and cycles
- every path/cycle alternates between edges from M, M'
- if $|M| < |M'|$, $\exists$ at least one path with start and ending edge in M'



Finding an augmenting path: $O(|V| \cdot |E|)$

State of the art matching: $O(|E|^{1+o(1)})$ (constant) for bipartite graphs,
$O(|V|^{1/2} \cdot |E|)$ in general

Theorem (Berge): Every non-maximum matching has an augmenting path

Algorithm: (outputs maximum matching M)

Start with $M = \emptyset$.

repeat
- look for aug. path P ← in bip. graphs: $O(|V| \cdot |E|)$ (with BFS); in general: $O(|V| \cdot |E|)$ (Edmond's blossom algorithm)
- if it doesn't exist, return M
- else $M := M \oplus P$

Theorem (Hall):

A bip. graph $G = (A \dot\cup B, E)$ contains matching M with $|M| = |A|$ if and only if

$\forall X \subseteq A: |X| \leq |N(X)|$ → I'm leaving out the proof (which is inductive)

(X: any subset; $N(X)$: neighbors)

→ **Corollary (Frobenius):** for all k: every k-regular bip. graph contains a perfect matching. $\Rightarrow$ Graph is union of perfect matchings

→ **Theorem:** In 2^t-regular (also k-regular) bip. graphs, one can find a perfect matching in $O(|E|)$

→ In bip. graphs, one can find a perfect matching in $O(|V||E|)$

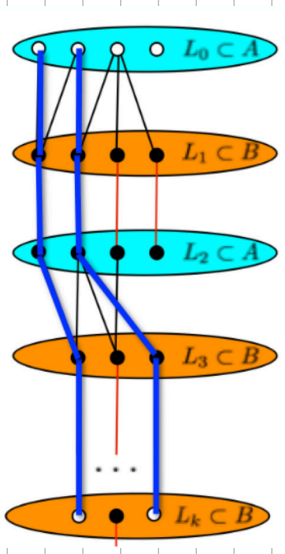
Matching algorithms:

Bip. graphs: Hopcroft-Karp (unweighted): $O(|V|^{1/2} \cdot |E|)$, $O(|E|^{1+o(1)})$ (constant) for polynomial weights

Generally (with polynomial weights): Micali-Vazirani (unweighted) / Gabow-Tarjani $O(|V|^{1/2} \cdot |E|)$, matrix mult. – Mucha, Sankowski (unweighted): $O(|V|^{2.373})$ (these guys have been trying to prove it for 30+ years)

BFS for alternating paths: Input: bip. graph, Matching M, Output: (shortest) aug. path if it exists

$L_0 := \{$ vertices (not covered by M) in $A\}$
Mark all vertices in L_0 as visited
for $i := 1$ to n
if i odd then
$L_i := \{$ non-visited neighbors of L_{i-1} through edges in $E \setminus M\}$
else
$L_i := \{$ non-visited neighbors of L_{i-1} through edges in $M\}$
if any v in L_i not covered
then return path to v (backtracking)



Inductively: $L_i = \{v \mid$ shortest alternating path from L_0 to v has length $i\}$

Hopcroft-Karp: Input: bip. graph, Output: maximum matching M
Start with $M = \emptyset$.
repeat until $\nexists$ aug. path
- $k_i :=$ length of shortest aug. path (can be found in $O(|V|+|E|)$)
- find multiple disjoint aug. paths of length k until we have a maximal set S from these paths (you cannot add an aug. path of length k to S)
- for all P from S:
  $M := M \oplus P$

Theorem: Hopcroft-Karp runs the repeat loop only $O(|V|^{1/2})$ times.
Its total runtime is $O(|V|^{1/2} \cdot (|V|+|E|))$.

A better approximation for metric TSP:

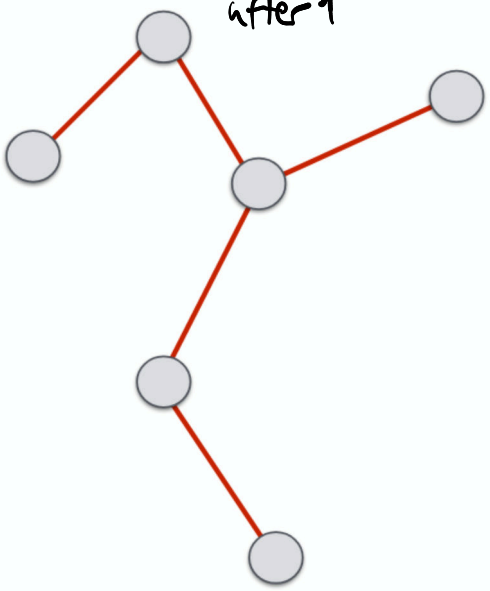
Theorem: Christofides' Algorithm computes a 1.5-approximation for the metric TSP in polynomial time (using MST, matching, and an Euler tour)
→ Karlin, Klein, Gharan found a $(3/2 - 10^{-36})$-approximation
→ 1.008-approx. is NP-complete

new steps:

1. find MST T ($\ell(T) \leq opt(K_n, \ell)$)
2'. $X :=$ vertices in T of odd degree, find min. matching M for X ($\ell(M) \leq \frac{1}{2} opt(K_n, \ell)$)
3. find Euler tour W ($\ell(W) = \ell(T) + \ell(M) \leq \frac{3}{2} opt(K_n, \ell)$)
4. go through W with shortcuts, s.t. every vertex visited once → Hamilton cycle C
   ($\ell(C) \leq \ell(W) = \ell(T) + \ell(M) \leq \frac{3}{2} opt(K_n, \ell)$)

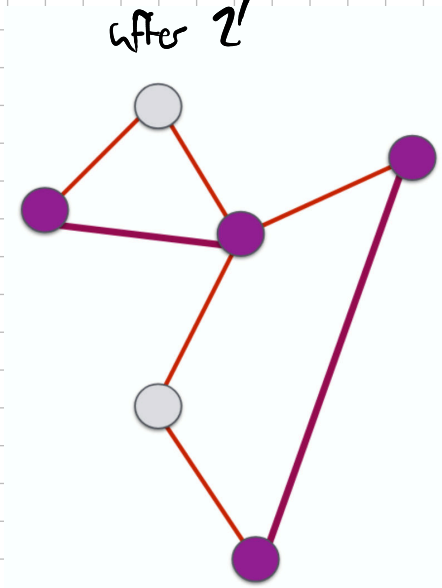
graphically:

after 1



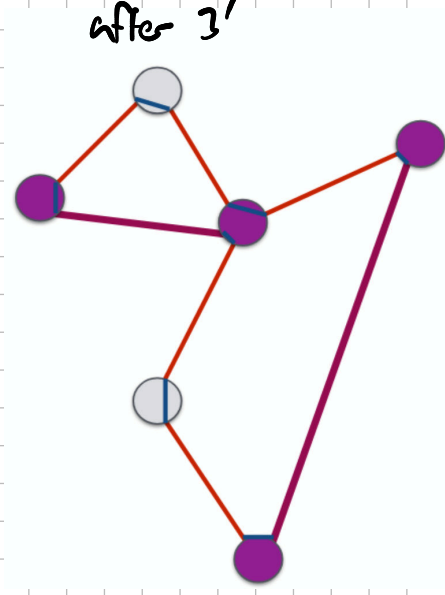
(2')

after 2'



(3)

after 3'



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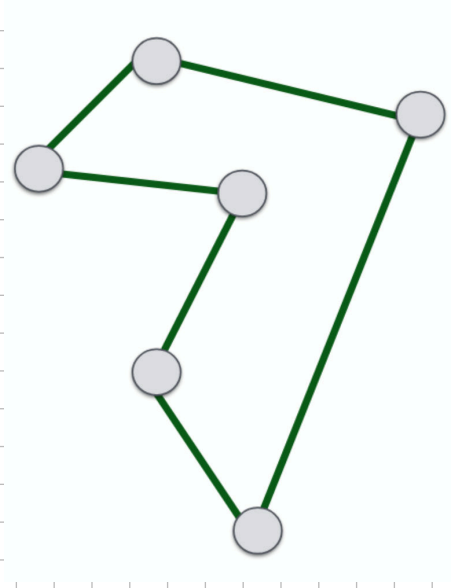


→ K-coloring: A coloring (of a graph) with k colors is a mapping $c: V \to [k]$, so that: $c(v) \neq c(v)$ for all edges $\{v, v\} \in E$.

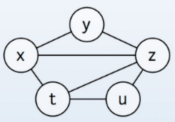
→ cool observation that I somehow didn't get in the lecture:

this reads like an adj. matrix

Live ranges

	x	y	z	t	u
x←1					
y←2					
z←x+y					
t←y					
u←x+t					
print z					
print t					
print u					

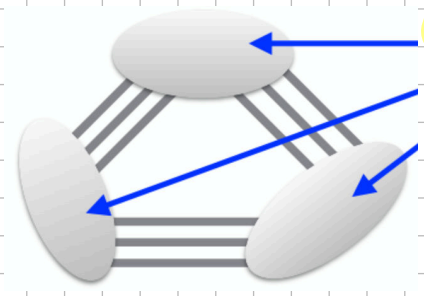
Interference graph



omg is this pronounced like ch in LaTeX? please get back to me on this one

→ Chromatic number $\chi(G)$ is min. amount of colors needed to color G.

→ $\chi(G) \leq k$ if and only if G k-partite



independent sets (no edges inside them)

→ special case: $\chi(G) \leq 2$ if and only if G bipartite

→ you can find this out in $O(|E|)$ with BFS/DFS

→ for $K \geq 3$, finding out if $\chi(G) \leq k$ is NP-complete

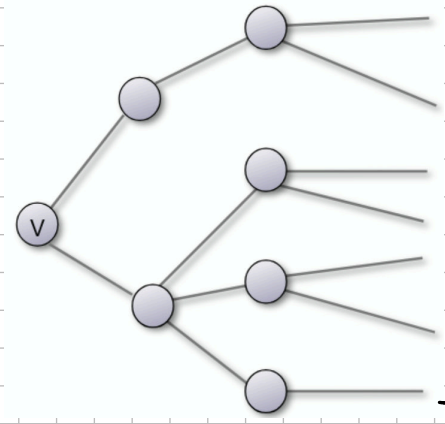
→ approximation not possible, $\forall \varepsilon > 0$ $n^{1-\varepsilon}$-approx. is NP-complete

→ exponential algorithm? yes (with sieve theorem (inclusion/exclusion)
→ polynomial space, time $O(2.2^n)$

→ special cases: yes

Theorem: $\forall k, r \in \mathbb{N}$: there are graphs without cycle of length $\leq k$, with $\chi(G) \geq r$

→ this practically looks like a tree



Greedy coloring (G):

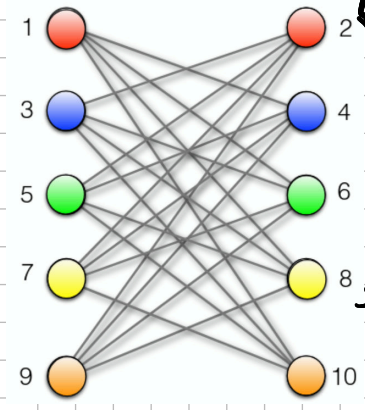
1. select arbitrary order of vertices: $V = \{v_1, \dots, v_n\}$
2. $c[v_1] \leftarrow 1$
3. for $i = 2$ to $i = n$ do
4. $c[v_i] \leftarrow \min\{k \in \mathbb{N} \mid k \neq c(u)$ for all $u \in N(v_i) \cap \{v_1, \dots, v_{i-1}\}\}$

(neighbors that came before i)

→ Observation: for all orders $V = \{v_1, \dots, v_n\}$ the greedy algo. needs at most $\Delta(G)+1$ colors ($\Delta(G) :=$ max. degree in G)

→ Observation: There is an order $V = \{v_1, \dots, v_n\}$, for which the greedy algo. only needs $\chi(G)$ colors.

→ Observation: There are bip. graphs and an order $V = \{v_1, \dots, v_n\}$, for which the greedy alg. needs $|V|/2$ colors.



→ Observation: If for the chosen order, $|N(v_i) \cap \{v_1, \dots, v_{i-1}\}| \leq k$ $\forall 2 \leq i \leq n$ holds, the greedy algo. maximally needs $k+1$ colors

→ Heuristic: V_n = vertices of the smallest degree. delete V_n.
V_{n-1} = vertices of smallest deg. in leftover graph. delete V_{n-1}.
Repeat.

→ If in every subgraph of G, there is a vertex of degree $\leq k$, the heuristic gives us an order $\{v_1, \dots, v_n\}$ for which the greedy algo. needs at most $k+1$ colors.

→ Corollary: The heuristic always finds a 2-coloring for trees

→ Theorem: If a graph is planar (can literally be drawn without edges intersecting), there is always a vertex of degree ≤ 5.

→ Corollary: The heuristic always finds a ≤ 6-coloring for planar graphs.

→ Corollary: G is connected and $\exists v \in V$ s.t. $\deg(v) < \Delta(G)$: the heuristic or DFS/BFS finds an order s.t. the greedy algo needs at most $\Delta(G)$ colors

← we can swap classes in graphs with independent sets

← this is used in block graphs;

Presume we can color every block with k colors.

Then, we can color G with k colors.

→ every graph can be colored in $O(|E|)$ with $\Delta(G)+1$ colors

→ Brooks' Theorem: $G \neq K_n$ (k-regular), $G \neq C_{2n+1}$ (complete), G connected:

⇒ G can be colored in $O(|E|)$ with $\Delta(G)$ colors.

Algorithm (based on Brooks' Theorem):

→ if $\Delta(G) = 2$ → color G with 2 colors

→ if $\exists v \in V$ with $\deg(v) < \Delta(G)$: color G with greedy algo. + heuristic

→ if $\exists$ cut vertex v: color all blocks (incl. v) with heuristic (+ color swap), s.t. v colored one certain way in all graphs

→ determine vertices $v, x, y \in V$ with $x, y \in N(v)$ and $\{x, y\} \notin E$ (as G conn., not complete)

→ Consider $G' := G[V \setminus \{x, y\}]$:

→ if G' connected: use greedy, first x and y, then heuristic for G'

→ else ...

Theorem: A 3-colorable graph can be colored in time $O(|V| + |E|)$ using $O(\sqrt{|V|})$ colors.

Algorithm for this:

While $\exists$ a vertex v with $> \sqrt{|V|}$ uncolored neighbors:

Color v with new color and its neighbors with 2 new colors

This works because the graph is tripartite, so each neighborhood $N(v)$ is bip.

Delete all colored vertices. The remaining graph has max. degree $\Delta \leq \sqrt{|V|}$.

Color the remaining vertices using the greedy algorithm with $\Delta + 1$ new colors.

Probability theory (and the start of WS-W6):

Def.: discrete probability space: determined by the set of events

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$\Omega = \{\omega_1, \omega_2, \ldots\}$ of elementary events ω_i (this is a "lowercase" Ω). Every elementary event has a probability $Pr[\omega_i]$ with $0 \leq Pr[\omega_i] \leq 1$ and $\sum_{\omega \in \Omega} Pr[\omega] = 1$.

Def.: A subset $E \subseteq \Omega$ is an **event**. The probability $Pr[E]$ of an event is determined as $Pr[E] = \sum_{\omega \in E} Pr[\omega]$. $\overline{E} := \Omega \setminus E$ is defined as the complementary event to E.

Lemma: For events $A, B \subseteq \Omega$:

- $Pr[\emptyset] = 0$ and $Pr[\Omega] = 1$.
- $0 \leq Pr[A] \leq 1$.
- $Pr[\overline{A}] = 1 - Pr[A]$
- If $A \subseteq B$, $Pr[A] \leq Pr[B]$

Laplace space: A finite prob. (probability) space, where all elementary events have the same probability (like picking a card, any card)

Observation: In a Laplace space Ω, for every event E, the following holds:
$Pr[E] = \frac{|E|}{|\Omega|}$.

Warning: Combinatorics will reappear and be treated as a prerequisite in this course. Here is a non-trivial example from the lecture:
Consider the deck of cards C with $|C| = 52$. $\Omega = \{(X,Y) : X, Y \subseteq C, X \cap Y = \emptyset, |X| = |Y| = 5\}$. This is meant to emulate two players drawing 5 cards each.

Multiple Choice: Which descriptions of $|\Omega|$ are correct?
a) $52^5 \cdot 52^5$, b) $\binom{52}{10} \cdot \binom{10}{5}$, c) $\frac{52!}{5! \cdot 5! \cdot 42!}$, d) $\binom{52}{5} \cdot \binom{47}{5}$

→ a) is wrong, as 5 cards are removed from the "pool" of eligible cards after the first draw.
→ b) is correct: First, you take 10 cards out of 52, then you decide how sets of 5 cards are divided between the players.
→ d) is correct: remove 5 cards from the deck, then 5 from the reduced deck.
→ c) is correct: $\binom{52}{5} \cdot \binom{47}{5} \overset{\text{def. combination}}{=} \left(\frac{52!}{5!47!}\right)\left(\frac{47!}{5!42!}\right) = \frac{52!}{5!\cancel{47!}} \cdot \frac{\cancel{47!}}{5!42!} = \frac{52!}{5!5!42!}$

Union of events:

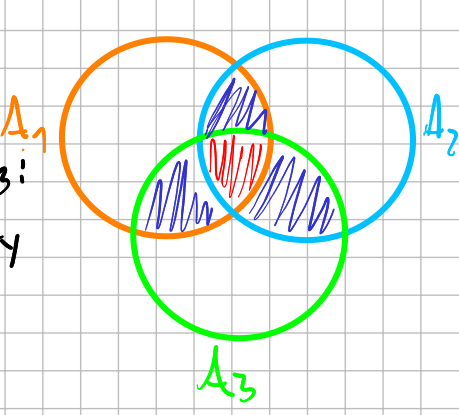
Addition theorem: for pairwise disjoint events $A_1, \ldots, A_n$: $Pr\left[\dot{\bigcup_{i=1}^{n}} A_i\right] = \sum_{i=1}^{n} Pr[A_i]$

Union Bound (Theorem): For events $A_1, \ldots, A_n$: $Pr\left[\bigcup_{i=1}^{n} A_i\right] \leq \sum_{i=1}^{n} Pr[A_i]$

Proof (optional, I found it interesting): For $i \in \{1, \ldots, n\}$: $B_i = A_i \setminus (A_1 \cup A_2 \cup \ldots \cup A_{n-1})$.
Then: $Pr[B_i] \leq Pr[A_i]$ and for $i \neq j$, B_i and B_j are disjoint.

$$\Pr\left[\bigcup_{i=1}^{n} A_i\right] = \Pr\left[\bigcup_{i=1}^{n} B_i\right] = \sum_{i=1}^{n} \Pr[B_i] \overset{\text{subset}}{\leq} \sum_{i=1}^{n} \Pr[A_i]$$

↑ Addition theorem



Inclusion/Exclusion Theorem: **Intuition:** Consider events A_1, A_2, A_3:

→ How do we avoid double counting and just get the probability of their disjoint union?

$$\Pr[A_1 \cup A_2 \cup A_3] = \Pr[A_1] + \Pr[A_2] + \Pr[A_3] - \Pr[A_1 \cap A_2] - \Pr[A_1 \cap A_3] - \Pr[A_2 \cap A_3] + \Pr[A_1 \cap A_2 \cap A_3]$$

Theorem: For events $A_1, \dots, A_n$ $(n \geq 2)$: $\Pr\left[\bigcup_{i=1}^{n} A_i\right] = \sum_{l=1}^{n} (-1)^{l+1} \cdot \sum_{1 \leq i_1 < \dots < i_l \leq n} \Pr[A_{i_1} \cap \dots \cap A_{i_l}]$

$= \sum_{i=1}^{n} \Pr[A_i] - \sum_{1 \leq i_1 < i_2 \leq n} \Pr[A_{i_1} \cap A_{i_2}] + \dots - \dots + \dots + (-1)^{n+1} \cdot \Pr[A_1 \cap \dots \cap A_n]$.

Example: Consider a deck of n cards: $A :=$ "There is ~~a light and it never goes out~~ a card that is in the same position after mixing the deck." Determine $\Pr[A]$.

Idea: $A_i =$ "Card i in the same position as before mixing" $\Rightarrow A = \bigcup_{i=1}^{n} A_i \Rightarrow$ inclusion/exclusion

For $S \subseteq [n]$: $A_S =$ "A holds for all cards in S" $= \bigcap_{i \in S} A_i \Rightarrow \Pr[A_S] = \frac{(n-|S|)!}{n!}$

Explaining $\frac{(n-|S|)!}{n!}$: ← we fix $|S|$ cards in their positions, only $n-|S|$ cards can move ← number of outcomes, all permutations are equally likely

Using inclusion/exclusion:

$$\Pr\left[\bigcup_{i=1}^{n} A_i\right] = \sum_{i=1}^{n} \Pr[A_i] - \sum_{S \subseteq [n], |S|=2} \Pr[A_S] + \sum_{S \subseteq [n], |S|=3} \Pr[A_S] - \dots$$

$$= n \cdot \frac{(n-1)!}{n!} - \binom{n}{2} \cdot \frac{(n-2)!}{n!} + \binom{n}{3} \cdot \frac{(n-3)!}{n!} - \dots$$

$$= 1 - \frac{1}{2!} + \frac{1}{3!} - \dots \pm \frac{1}{n!} \overset{n \to \infty}{\to} 1 - \frac{1}{e} \approx 63{,}2\%$$

$$\binom{n}{k} \cdot \frac{(n-k)!}{n!} = \frac{n!}{k!(n-k)!} \cdot \frac{(n-k)!}{n!} = \frac{1}{k!}$$

Conditional probabilities:

Def.: Let A and B be events with $\Pr[B] > 0$. The cond. (conditional) prob. $\Pr[A|B]$ of A given B is defined as $\Pr[A|B] := \frac{\Pr[A \cap B]}{\Pr[B]}$.

The following hold: $\Pr[A|\Omega] = \Pr[A]$, $\Pr[A|A] = 1$, $\Pr[A|\bar{A}] = 0$,

$\Pr[A \cap B] = \Pr[A|B] \cdot \Pr[B] = \Pr[B|A] \cdot \Pr[A]$

Multiplication theorem: Let $A_1, \dots, A_n$ be events. If $\Pr[A_1 \cap \dots \cap A_n] > 0$,

$\Pr[A_1 \cap \dots \cap A_n] = \Pr[A_1] \cdot \Pr[A_2|A_1] \cdot \Pr[A_3|A_1 \cap A_2] \cdots \Pr[A_n|A_1 \cap \dots \cap A_{n-1}]$.

Proof (also optional): $\Pr[A_1] \geq \Pr[A_1 \cap A_2] \geq \dots \geq \Pr[A_1 \cap \dots \cap A_n] > 0$

$\Rightarrow$ all probs. well-defined $\Rightarrow$ apply def. cond. prob.

$\Pr[A_1] \cdot \frac{\Pr[A_1 \cap A_2]}{\Pr[A_1]} \cdot \frac{\Pr[A_1 \cap A_2 \cap A_3]}{\Pr[A_1 \cap A_2]} \cdot \dots \cdot \frac{\Pr[A_1 \cap \dots \cap A_n]}{\Pr[A_1 \cap \dots \cap A_{n-1}]}$ (telescoping product!)

Non-trivial application: Birthday problem:

Consider a room of n people. Determine the probability of 2 people sharing a birthday.

Reshape this: we have n balls and n bins. Here, $n = 365$. Determine the prob. of 2 balls landing in the same bin () / the prob. of all balls landing in different bins.

Event $A_i =$ "i-th ball lands in a "free" bin". $\rightarrow \Pr[A_1] = 1$, $\Pr[A_2|A_1] = \frac{n-1}{n} = 1 - \frac{1}{n}$,

$\Pr[A_i | A_1 \cap \dots \cap A_{i-1}] = \frac{n-(i-1)}{n} = 1 - \frac{i-1}{n}$

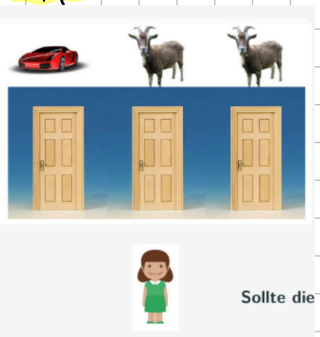
Mult. theorem: $\Pr[A_1 \cap \dots \cap A_n] = 1 \cdot (1 - \frac{1}{n}) \cdot (1 - \frac{2}{n}) \cdot \dots \cdot (1 - \frac{(n-1)}{n})$

Estimating the products: $1 - x \approx e^{-x}$ if x small (more accurately: $1 - x \leq e^{-x(1-O(x))}$ ($O(x)$: a constant), $1 - x \leq e^{x}$ $\forall x \in \mathbb{R}$)

$(1 - \frac{1}{n})(1 - \frac{2}{n}) \cdots (1 - \frac{n-1}{n}) \approx e^{-\frac{1}{n}} \cdot e^{-\frac{2}{n}} \cdot \dots \cdot e^{-\frac{n-1}{n}} \approx e^{-\frac{1}{n}(1+2+\dots+(n-1))} \approx e^{-\frac{(n-1)\cdot n}{2n}}$

Theorem of total probability: Let $A_1, \dots, A_n$ be pairwise disjoint events and let $B \subseteq A_1 \cup \dots \cup A_n$. Then, $\Pr[B] = \sum_{i=1}^{n} \Pr[B|A_i] \cdot \Pr[A_i]$.

Application (goat problem):



There are 3 doors: there is a goat behind 2 of them and a car behind 1. The candidate chooses a door and the host opens a door that has a goat behind it (not the one chosen). The candidate is offered the chance to change doors, should she do it?

Using the theorem: $A_1 =$ "Candidate chose the correct door in the beginning",

$A_2 =$ "She chose the wrong door", $B =$ "She wins if she switches doors"

Keep in mind that by the theorem, $\Pr[B] = \sum_{i=1}^{n} \Pr[B|A_i] \cdot \Pr[A_i]$.

$\Pr[B|A_1] = 0$, $\Pr[B|A_2] = 1$, $\Pr[A_1] = \frac{1}{3}$, $\Pr[A_2] = \frac{2}{3}$. $\Rightarrow \Pr[B] = 0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = \frac{2}{3}$.

Bayes' Theorem: Let $A_1, \dots, A_n$ be disjoint events and $B \subseteq A_1 \cup \dots \cup A_n$ be an event with $\Pr[B] > 0$. Then, $\forall i \in \{1, \dots, n\}$ $\Pr[A_i|B] = \frac{\Pr[B|A_i] \cdot \Pr[A_i]}{\sum_{j=1}^{n} \Pr[B|A_j] \cdot \Pr[A_j]}$

Proof: $\Pr[B|A_i] \cdot \Pr[A_i] = \Pr[B \cap A_i]$, $\sum_{j=1}^{n} \Pr[B|A_j] \cdot \Pr[A_j] = \Pr[B]$ (Th. of total prob.) $\Rightarrow$ true by def. of cond. prob.

Example: Test for a disease (we had this exact one at school)

A_1 = „you have the disease", A_2 = „you don't have it", B = „Test positive"

Known: Sensitivity → $Pr[B|A_1] = 72\%$, Specificity → $Pr[B|A_2] = 98\%$

$$Pr[A_1|B] = \frac{Pr[B|A_1] \cdot Pr[A_1]}{Pr[B|A_1] \cdot Pr[A_1] + Pr[B|A_2] \cdot Pr[A_2]} = \frac{0.72 \cdot Pr[A_1]}{0.72 \cdot Pr[A_1] + 0.02 \cdot Pr[A_2]} \leadsto \begin{cases} 26.7\% \text{ if } Pr[A_1] = 1\% \\ 0.72\% \text{ if } Pr[A_1] = 0.02\% \end{cases}$$

Independence:

intuitively: $Pr[A|B]$ gives $Pr[A]$ under the condition of B. A and B are independent, if B coming into effect does not change $Pr[A]$, so $\frac{Pr[A \cap B]}{Pr[B]} = Pr[A|B] = Pr[A]$

Def.: Events A and B are independent if $Pr[A \cap B] = Pr[A] \cdot Pr[B]$

Example: Roll 2 dice. $A :=$ „Sum of face values is 7", $B :=$ „The second die's face value is 2"

$Pr[A] = \frac{6}{36}$, $Pr[B] = \frac{1}{6}$, $Pr[A \cap B] = \frac{1}{36} = Pr[A] \cdot Pr[B] \Rightarrow$ independent

Independence of multiple events:

Def.: Events $A_1, \dots, A_n$ are independent if for all subsets $J \subseteq \{1, \dots, n\}$, $Pr[\bigcap_{j \in J} A_j] = \prod_{j \in J} Pr[A_j]$ (1)

An infinite group of events A_i with $i \in \mathbb{N}$ is independent if (1) holds for all finite subsets $J \subseteq \mathbb{N}$.

Lemma: Let $A_1, \dots, A_n$ be events, A_i^0 be $\bar{A}_i$ and A_i^1 be A_i ($\forall i \in \{1, \dots, n\}$). Then, $A_1, \dots, A_n$ are independent if and only if for all $(s_1, \dots, s_n) \in \{0,1\}^n$: $Pr[\bigcap_{j=1}^{n} A_j^{s_j}] = \prod_{j=1}^{n} Pr[A_j^{s_j}]$

→ The proof is omitted to save space and time, but it is available in the Lecture 11 notes on Moodle and worth understanding

Misconception: $A_1, \dots, A_n$ are independent if $Pr[A_1, \dots, A_n] = \prod_{i=1}^{n} Pr[A_i]$ ← this is false!

Lemma: Let A, B, C be independent. Then, 1) $A \cap B$ and C are independent, 2) $A \cup B$ and C are independent.

Proof of the lemma: 1) $Pr[(A \cap B) \cap C] = Pr[A] \cdot Pr[B] \cdot Pr[C] = Pr[A \cap B] \cdot Pr[C]$

2) $Pr[(A \cup B) \cap C] = Pr[(A \cap C) \cup (B \cap C)] \overset{\text{sieve formula}}{=} Pr[A \cap C] + Pr[B \cap C] - Pr[A \cap B \cap C]$

$\overset{\text{ind. of } A,B,C}{=} Pr[C] \cdot (\overbrace{Pr[A] + Pr[B] - Pr[A \cap B]}^{= Pr[A \cup B], \text{ sieve formula}})$

Random variables:

Def.: A random variable is a map $X: \Omega \to \mathbb{R}$, with Ω being the outcome range of a probability space.

Example: Dice → $\Omega = \{1,2,3,4,5,6\}$

$X(\omega) = \begin{cases} 1 & \text{if } \omega \text{ prime} \\ 0 & \text{else} \end{cases}$ (Description as Bernoulli trial), or $X(\omega) = \omega^2 - 5\omega$ (Polynomial description)

Note: The polynomial does not have to be meaningful, it just needs to be a function

Def.: the range of a random variable X is $\Omega_X := X(\Omega) \{\alpha \in \mathbb{R} \mid \exists \omega \in \Omega \text{ with } X(\omega) = \alpha\}$

→ Range is defined the same as for functions in Calculus → set of all possible values of X

Notation: „$X \leq 5$" describes the event $\{\omega \in \Omega : X(\omega) \leq 5\}$ → the event that X "returns" a value ≤ 5.

[I got carried away, W7-W8 starts here]

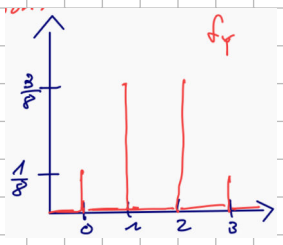
Probability Density Function (PDF): $f_X : \mathbb{R} \to [0,1]$, $\alpha \to \Pr[X = \alpha]$

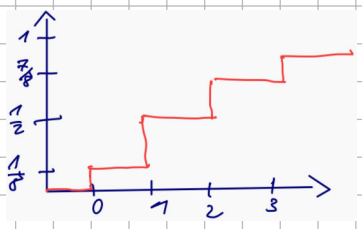
Cumulative Distribution Function (CDF): $F_X : \mathbb{R} \to [0,1]$, $\alpha \to \Pr[X \leq \alpha]$

Example: Roll a die thrice. Probability Space $\Omega = \{1, \dots, 6\}^3$

Random var. Y = amount of odd face values rolled

$\Pr[Y=0] = \frac{1}{8}$, $\Pr[Y=1] = \frac{3}{8}$, $\Pr[Y=2] = \frac{3}{8}$, $\Pr[Y=3] = \frac{1}{8}$

PDF: 

($f_Y(0) = \frac{1}{8}, \dots$) CDF: 

($F_Y(1) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$)

Expected value:

Def.: for a random variable X with range W_X, the expected value $\mathbb{E}[X]$ is defined as $\mathbb{E}[X] := \sum_{\alpha \in W_X} \alpha \cdot \Pr[X = \alpha]$ if the sum converges absolutely. Else, the expected value is undefined. (This case will not be applied in this course)

Lemma: for a random variable X, $\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) \cdot \Pr[\omega]$

→ this is near equivalent to the def.

Special case $W_X \subseteq \mathbb{N}_0$:

Theorem: Let X be a random variable with $W_X \subseteq \mathbb{N}_0$. Then: $\mathbb{E}[X] = \sum_{i=1} \Pr[X \geq i]$

Proof: $\mathbb{E}[X] = \sum_{i=0}^{\infty} i \cdot \Pr[X = i] = \underbrace{\sum_{i=0}^{\infty} \sum_{j=1}^{i} \Pr[X = i]}_{\text{sum over pairs } (i,j) \text{ with } i \geq j \geq 1} = \sum_{j=1}^{\infty} \sum_{i=j}^{\infty} \Pr[X = i] = \sum_{j=1}^{\infty} \Pr[X \geq j]$

Example (typical): Toss a coin with p[„Heads"] $= p$ $(0 < p < 1)$, X = amount of throws until first „Heads". $\mathbb{E}[X] = \sum_{i=1}^{\infty} \underbrace{\Pr[X \geq i]}_{\Pr[\text{„first } i-1 \text{ tosses are Tails"}]} = \sum_{i=1}^{\infty} (1-p)^{i-1} = \sum_{i=0}^{\infty} (1-p)^i = \frac{1}{p}$

Indicator variables: For an event $A \subseteq \Omega$, the indicator variable (ind. var.) X_A is defined as

$$X_A(\omega) := \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{else} \end{cases}$$

Observation: for the ind. var. X_A of an event A: $\mathbb{E}[X_A] = \Pr[A]$

Linearity of the expected value:

Theorem: for random variables $X_1, \dots, X_n$ and $X := a_1 \cdot X_1 + \dots + a_n \cdot X_n + b$ with $a_1, \dots, a_n, b \in \mathbb{R}$: $\mathbb{E}[X] := a_1 \cdot \mathbb{E}[X_1] + \dots + a_n \cdot \mathbb{E}[X_n] + b$

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→ „$\mathbb{E}$ of the sum = sum of expected values"

Proof: $\mathbb{E}[X] = \sum_{w\in\Omega} X(w)\cdot Pr[w] = \sum_{w\in\Omega}(a_1\cdot X_1(w) + \dots + a_n\cdot X_n(w) + b)\cdot Pr[w]$

$= a_1\cdot \underbrace{(\sum_{w\in\Omega} X_1(w)\cdot Pr[w])}_{\mathbb{E}[X_1]} + \dots + a_n\underbrace{(\sum_{w\in\Omega} X_n(w)\cdot Pr[w])}_{\mathbb{E}[X_n]} + b\cdot \underbrace{(\sum_{w\in\Omega} Pr[w])}_{1}$

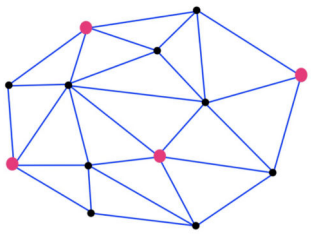
Example: We toss a coin thrice. $X :=$ amount of partial sequences that consist of 3 times heads, $\mathbb{E}[X]$?

$X = X_1 + \dots + X_{n-2}$ with $X_i := \begin{cases} 1 & \text{if at position } i \text{, the partial sequence HHH begins} \\ 0 & \text{else} \end{cases}$

$\mathbb{E}[X_i] = Pr[X_i = 1] = \frac{1}{8} \Rightarrow \mathbb{E}[X] = \sum_{i=1}^{n-2}\mathbb{E}[X_i] = \frac{n-2}{8}$

↑ Linearity of $\mathbb{E}$

Practical Use: Stable Set



Networking: each vertex is a computer, we want to find the maximum stable set

→ stable set: vertices not connected by edges

Goal: Every computer v computes $S_v \in \{0,1\}$ s.t. $S = \{v\in V : S_v = 1\}$ is a stable set.

Algorithm:

① Set $S_v \leftarrow 1$ with probability p and $S_v \leftarrow 0$ else

② For all $v\in V$ with $\{v,v\}\in E$: } Information exchange between neighbors

If: $S_v = S_v = 1$: Set one of S_v and S_v to 0

↑ guarantees correctness

③ Return : S_v

Theorem: The algorithm computes a stable set S with $\mathbb{E}[|S|] \geq n\cdot p - m\cdot p^2$ ($n = |V|$, $m = |E|$)

$X :=$ amount of vertices v with $S_v = 1$ following ①, $Y =$ amount of edges considered in ②

$e = \{v,v\}$ is considered if $S_v = S_v = 1$ after ①

Observation: $|S| \geq X - Y$

Linearity of $\mathbb{E}$: $\Rightarrow \mathbb{E}[|S|] \geq \mathbb{E}[X] - \mathbb{E}[Y]$

Determine $\mathbb{E}[X]$: $X_v :=$ ind.var. for „$S_v = 1$ after ①"

Then: $X = \sum_{v\in V} X_v \Rightarrow \mathbb{E}[X] = \sum_{v\in V}\underbrace{\mathbb{E}[X_v]}_{(=p)} = n\cdot p$

$\mathbb{E}[Y]$: Y_e ( ?) := ind.var. for „edge e considered in ②"

Then: $Y = \sum_{e\in E} Y_e \Rightarrow \mathbb{E}[Y] = \sum_{e\in E}\underbrace{\mathbb{E}[Y_e]}_{= Pr[e \text{ considered in } ②]} = m\cdot p^2$

$e = \{v,v\} = Pr[S_v \geq 1 \text{ and } S_v = 1 \text{ after } ①]$

$= Pr[S_v = 1 \text{ after } ①]\cdot Pr[S_v = 1 \text{ after } ①]$

$= p\cdot p$

Optimal choice of p: $p = \frac{n}{2m}$, Then $\mathbb{E}[|S|] \geq \frac{n^2}{4m} \geq \frac{n}{2k}$ ($m = \frac{k\cdot n}{2}$)

↑ for k-regular graphs

Variance:

Def.: For a random variable X with $\mu = \mathbb{E}[X]$, the variance $\mathrm{Var}[X]$ is defined as:

$\mathrm{Var}[X] := \mathbb{E}[(X-\mu)^2]$ with $\sigma = \sqrt{\mathrm{Var}[X]}$ is the standard deviation of X.

Example: $Pr[X=-10^{10}] = \frac{1}{2} = Pr[X=10^{10}] \Rightarrow \mu = 0, \mathrm{Var}[X] = 10^{20}, \sigma = 10^{10}$

Theorem: For any random variable X, $\mathrm{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

Proof: $\mathrm{Var}[X] = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \overset{\text{lin. of } \mathbb{E}}{=} \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \overset{\mu = \mathbb{E}[X]}{=} \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

Theorem: For any random variable X and $a, b \in \mathbb{R}$: $\mathrm{Var}[a \cdot X + b] = a^2 \cdot \mathrm{Var}[X]$.

Proof: ① $\mathrm{Var}[X+b] = \mathrm{Var}[X]$: Lin. of $\mathbb{E}$: $\mathbb{E}[X+b] = \mathbb{E}[X] + b \Rightarrow \mathrm{Var}[X+b] = \mathbb{E}[(X+b - \mathbb{E}[X+b])^2]$

$= \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathrm{Var}[X]$

② $\mathrm{Var}[a \cdot X] = a^2 \cdot \mathrm{Var}[X]$: $\mathrm{Var}[a \cdot X] \overset{\text{↑Theorem}}{=} \mathbb{E}[(aX)^2] - \mathbb{E}[a \cdot X]^2 = a^2\mathbb{E}[X^2] - (a \cdot \mathbb{E}[X])^2 \overset{\text{Theorem}}{=} a^2 \cdot \mathrm{Var}[X]$

Moments:

Def.: For a random variable X, $\mathbb{E}[X^k]$ is the k-th moment and $\mathbb{E}[(X - \mathbb{E}[X])^k]$ is the k-th central moment.

First moment: $\mathbb{E}[X]$, **first central moment** $= \mathbb{E}[X] - \mathbb{E}[X] = 0$, **2nd central moment:** $\mathbb{E}[(X - \mathbb{E}[X])^2]$

$= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mathrm{Var}[X]$

Important Distributions:

Def.: A random variable X with range $W_X = \{0,1\}$ and PDF $f_X(x) = \begin{cases} p & \text{for } x = 1 \\ 1-p & \text{for } x = 0 \\ 0 & \text{else} \end{cases}$

is **Bernoulli distributed** with probability of success $= p$, which is written as $X \sim \mathrm{Bernoulli}(p)$

→ this describes binary events, such as various ind. vars.

→ $\mathbb{E}[X] = p$, $\mathrm{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = p - p^2 = p(1-p)$

Def.: A random variable X with range $W_X = \{0, \dots, n\}$ and PDF $f_X(x) = \begin{cases} \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k} & \text{for } k \in \{0, \dots, n\} \\ 0 & \text{else} \end{cases}$

is **binomially distributed** with parameters p and n, written $X \sim \mathrm{Bin}(n, p)$

Example: Toss a coin n times, with $Pr[\text{"Heads"}] = p$, $Pr[\text{"Tails"}] = 1-p$, $X :=$ amount of heads thrown

$Pr[X=k] = \binom{n}{k} p^k (1-p)^{n-k}$ $\Rightarrow$ n positions, depending on results of n tosses $\Rightarrow$ choose k positions that are heads

$\Rightarrow \binom{n}{k}$ possibilities

Prob. that n throws have exactly this result: $p^k(1-p)^{n-k}$

Def.: A random variable with PDF $f_X(i) = \begin{cases} \frac{e^{-\lambda}\lambda^i}{i!} & \text{for } i \in \mathbb{N}_0 \\ 0 & \text{else} \end{cases}$ is **Poisson distributed** with parameter λ,

written as $X \sim Po(\lambda)$

→ $\mathrm{Bin}(n, \frac{\lambda}{n})$ converges to $Po(\lambda)$ for $n \to \infty$, $\mathbb{E}[X] = \mathrm{Var}[X] = \lambda$

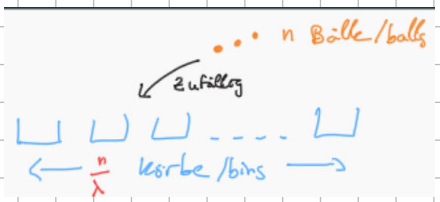
Proof of these observations:

$X :=$ amount of balls in first bin $\sim Bin(n, \frac{\lambda}{n})$

$\Rightarrow \mathbb{E}[X] = n \cdot \frac{\lambda}{n} = \lambda$ $\quad \sim (e^{-\frac{\lambda}{n}})^{n-i} = e^{-\lambda \frac{n-i}{n}} = e^{-\lambda}$

$\lim_{n\to\infty} Pr[X=i] = \lim_{n\to\infty} \binom{n}{i} (\frac{\lambda}{n})^i \underbrace{(1-\frac{\lambda}{n})^{n-i}}_{\sim} = \frac{\lambda^i}{i!} e^{-\lambda}$

$= \frac{\lambda^i}{i!} (\frac{n}{n} \cdot \frac{(n-1)}{n} \cdot \ldots \cdot \frac{(n-i+1)}{n})$



Use case: Describing rare events, such as "$X =$ amount of heart attacks in 🇨🇭 in the next hour"

Def.: A random variable X with PDF $f_X(i) = \begin{cases} p \cdot (1-p)^{i-1} & \text{for } i \in \mathbb{N} \\ 0 & \text{else} \end{cases}$ is geometrically distributed with $p =$ probability of success, written as $X \sim Geo(p)$

$\rightarrow \mathbb{E}[X] = \frac{1}{p}$, $Var[X] = \frac{1-p}{p^2}$, CDF: $F_X(n) = Pr[X \le n] = \sum_{i=1}^{n} Pr[X=i] = \sum_{i=1}^{n} p \cdot (1-p)^{i-1} = 1 - (1-p)^n$

Example: Toss a coin n times with $Pr[\text{"Heads"}] = p$, $Pr[\text{"Tails"}] = 1-p$ until we have gotten heads once,

$X :=$ Amount of tosses.

$Pr[X=k] = \underbrace{(1-p)^{k-1} \cdot p}_{\text{Tails } k-1 \text{ times, then heads}}$

Theorem: If $X \sim Geo(p)$, $\forall s,t \in \mathbb{N}$: $Pr[X \ge s+t \mid X > s] = Pr[X \ge t]$

Def.: For $n \ge 2$, a random variable X with density $f_X(k) = \begin{cases} \binom{k-1}{n-1} \cdot p^n \cdot (1-p)^{k-n} & \text{for } k = 1, 2, \ldots \\ 0 & \text{else} \end{cases}$

is negatively binomial distributed with order n

$\rightarrow \mathbb{E}[X] = \frac{n}{p}$

$X_i :=$ amount of coin tosses after $(i-1)$th success up to and including i-th success

Then: $X_i \sim Geo(p)$, $X = \sum_{i=1}^{n} \Rightarrow \mathbb{E}[X] = \sum_{i=1}^{n} \underbrace{\mathbb{E}[X_i]}_{\frac{1}{p}} = n \cdot \frac{1}{p} = \frac{n}{p}$

Example (waiting for n-th success): Toss a coin n times with $Pr[\text{"Heads"}] = p$, $Pr[\text{"Tails"}] = 1-p$ until we get heads n times, $X :=$ amount of tosses

$Pr[X=k] = \binom{k-1}{n-1} (1-p)^{k-n} \cdot p^n$

$\rightarrow$ k^{th} toss = heads, select $n-1$ other tosses that would have shown heads $\rightarrow \binom{k-1}{n-1}$ possibilities

$\rightarrow Pr[\text{"}k \text{ tosses have same result"}]: (1-p)^{k-n} \cdot p^n$

Multiple Random Variables:

Let X, Y be random variables:

joint PDF: $f_{X,Y}(\alpha, \beta) = Pr[X = \alpha, Y = \beta]$, **edge density of X (= PDF of X):** $f_X(\alpha) = \sum_{\beta \in W_Y} f_{X,Y}(\alpha, \beta)$

$= \sum_{\beta \in W_Y} Pr[X = \alpha, Y = \beta] = Pr[X = \alpha]$

Example: Cast a die. $X :=$ face value of die, after this, toss a coin X times. $Y :=$ amount of heads.

$f_{X,Y}(n,k) = Pr[X=n, Y=k] = \begin{cases} \frac{1}{6}(k) \cdot \frac{1}{2^n} & \text{if } n \in \{1,\ldots,6\}, k \in \{0,1,\ldots,n\} \\ 0 & \text{else} \end{cases}$

$Pr[Y=k] = f_Y(k) = \begin{cases} \sum_{n=\max\{1,k\}}^{6} \frac{1}{6}\binom{n}{k}\frac{1}{2^n} & \text{if } k \in \{0,1,\ldots,6\} \\ 0 & \text{else} \end{cases}$

Independence of random variables:

Def.: Random variables $X_1, \dots, X_n$ independent **if and only if** $\forall (a_1, \dots, a_n) \in W_{X_1} \times \dots \times W_{X_n}$:

$$\underbrace{Pr[X_1 = a_1, \dots, X_n = a_n]}_{f_{X_1,\dots,X_n}(a_1,\dots,a_n)} = \underbrace{Pr[X_1 = a_1]}_{f_{X_1}(a_1)} \cdot \dots \cdot \underbrace{Pr[X_n = a_n]}_{f_{X_n}(a_n)}$$

→ "joint PDF = product of edge densities"

Observation: Events $A_1, \dots, A_n$ are independent if and only if indicator variables $I_{A_1}, \dots, I_{A_n}$ are independent.

Lemma: Let $X_1, \dots, X_n$ be independent random variables and $S_1, \dots, S_n \subseteq \mathbb{R}$ any sets:

$$Pr[X_1 \in S_1, \dots, X_n \in S_n] = Pr[X_1 \in S_1] \cdot \dots \cdot Pr[X_n \in S_n]$$

Proof: Assume $S_i \subseteq W_{X_i}$ $\forall i \in [n]$, S_i countable:

$$Pr[X_1 \in S_1, \dots, X_n \in S_n] = \sum_{a_1 \in S_1} \dots \sum_{a_n \in S_n} Pr[X_1 = a_1, \dots, X_n = a_n]$$

$$\overset{\text{Ind. of } X_1,\dots,X_n}{=} \sum_{a_1 \in S_1} \dots \sum_{a_n \in S_n} Pr[X_1 = a_1] \cdot \dots \cdot Pr[X_n = a_n]$$

$$= \underbrace{\left(\sum_{a_1 \in S_1} Pr[X_1 = a_1]\right)}_{Pr[X_1 \in S_1]} \cdot \dots \cdot \underbrace{\left(\sum_{a_n \in S_n} Pr[X_n = a_n]\right)}_{Pr[X_n \in S_n]}$$

Corollary: Let $X_1, \dots, X_n$ be independent random variables and $I \subseteq [n]$. Then, the random variables $(X_i)_{i \in I}$ are also independent.

Proof: Let $I = \{i_1, \dots, i_k\}$ and $(a_{i_1}, \dots, a_{i_k}) \in W_{X_1} \times \dots \times W_{X_k}$.

For $i \in [n]$, let $S_i = \begin{cases} \{a_i\} & \text{if } i \in I \\ W_{X_i} & \text{else} \end{cases}$

Then, $Pr[X_{i_1} = a_{i_1}, \dots, X_{i_k} = a_{i_k}] = Pr[X_1 \in S_1, \dots, X_n \in S_n]$

$$\overset{\text{Lemma}}{=} Pr[X_1 \in S_1] \cdot \dots \cdot Pr[X_n \in S_n]$$

$$= Pr[X_{i_1} = a_{i_1}] \cdot \dots \cdot Pr[X_{i_k} = a_{i_k}]$$

Theorem: Let $f_1, \dots, f_n$ be functions with $f_i : \mathbb{R} \to \mathbb{R}$ for $i = 1, \dots, n$. If $X_1, \dots, X_n$ are independent, this holds for $f_1(X_1), \dots, f_n(X_n)$.

Proof: For $i = 1, \dots, n$ consider values $a_1, \dots, a_n$ with $a_i \in W_{f_i(X_i)}$

$S_i = \{b : f_i(b) = a_i\}$

Then: $Pr[f_1(X_1) = a_1, \dots, f_n(X_n) = a_n] = Pr[X_1 \in S_1, \dots, X_n \in S_n]$

$$\overset{\text{independence}}{=} Pr[X_1 \in S_1] \cdot \dots \cdot Pr[X_n \in S_n]$$

$$= Pr[f_1(X_1) = a_1] \cdot \dots \cdot Pr[f_n(X_n) = a_n]$$

Combined Random Variables:

Theorem: For two independent random variables X and Y, let $Z := X + Y$:

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$f_Z(\alpha) = \sum_{\beta \in W_X} f_X(\beta) \cdot f_Y(\alpha - \beta)$.

Proof: $f_Z(\alpha) = Pr[X+Y=\alpha] = \sum_{\beta \in W_X} Pr[X=\beta \text{ and } Y=\alpha-\beta]$

$\overset{\text{independence}}{=} \sum_{\beta \in W_X} \underbrace{Pr[X=\beta]}_{f_X(\beta)} \cdot \underbrace{Pr[Y=\alpha-\beta]}_{f_Y(\alpha-\beta)}$

Corollary: If $X \sim Po(\lambda)$, $Y \sim Po(\mu)$, X, Y are independent, then $X+Y \sim Po(\lambda+\mu)$

Moments of Combined Random Variables:

Theorem: For independent random variables $X_1, \dots, X_n$: $\mathbb{E}[X_1 \cdot \dots \cdot X_n] = \mathbb{E}[X_1] \cdot \dots \cdot \mathbb{E}[X_n]$.

Theorem: For independent random variables $X_1, \dots, X_n$: $Var[X_1 + \dots + X_n] = Var[X_1] + \dots + Var[X_n]$.

→ I have omitted the proofs from the lecture because they do not prove the general case.

Wald's Identity:

Theorem: Let N and X be two independent random variables with $W_N \subseteq \mathbb{N}$.

Furthermore, let $Z := \sum_{i=1}^{N} X_i$, with $X_1, X_2, \dots$ being independent copies of X.

Then, $\mathbb{E}[Z] = \mathbb{E}[N] \cdot \mathbb{E}[X]$.

Example: Cast a die, let $N :=$ face value of die. Then, throw a coin N times.

Let $Z :=$ amount of heads. What is $\mathbb{E}[Z]$? $\mathbb{E}[Z] = \underbrace{3.5}_{\mathbb{E}[N]} \cdot \underbrace{\frac{1}{2}}_{\mathbb{E}[\text{heads on a coin toss}]}$

Conditional Random Variables:

X = random variable, A = event with $Pr[A] > 0$.

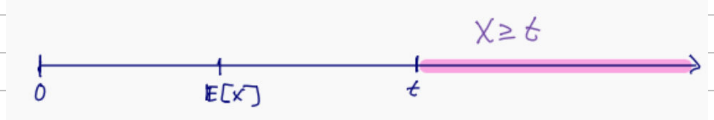
$X|A \Rightarrow$ Prob. of X taking on a value conditional on A.

$\Rightarrow Pr[(X|A) = \alpha] := Pr[X=\alpha|A]$, $\mathbb{E}[X|A] = \sum_{\alpha \in W_X} \alpha \cdot Pr[X=\alpha|A]$

Markov Inequality:

Theorem: Let X be a random var. that does not take on negative values. Then, for all $t \in \mathbb{R}$ with $t > 0$, $Pr[X \geq t] \leq \frac{\mathbb{E}[X]}{t}$, or equivalently: $Pr[X \geq t \cdot \mathbb{E}[X]] \leq \frac{1}{t}$

Diagram:



→ this is used to upper bound probabilities given $\mathbb{E}[X]$.

Proof: take X = non-neg. random var. / $t > 0$: (to show: $Pr[X \geq t] \leq \frac{\mathbb{E}[X]}{t}$)

$\mathbb{E}[X] = \sum_{\alpha \in W_X} \alpha \cdot Pr[X=\alpha]$

$= \sum_{\substack{\alpha \in W_X \\ \alpha < t}} \underbrace{\alpha \cdot Pr[X=\alpha]}_{\geq 0} + \sum_{\substack{\alpha \in W_X \\ \alpha \geq t}} \underbrace{\alpha}_{\geq t} \cdot Pr[X=\alpha]$

$\geq t \cdot \sum_{\substack{a \in \text{ran}X \\ a \geq t}} \Pr[X=a] = t \cdot \Pr[X \geq t]$ ∎

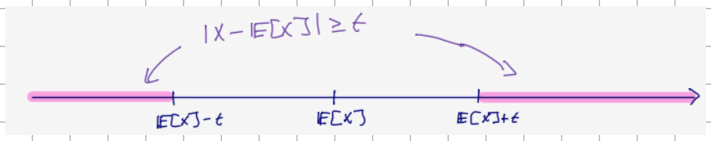
Chebyshev Inequality:

Theorem: Let X be a random variable and $t \in \mathbb{R}$ with $t > 0$.

Then, $\Pr[|X - \mathbb{E}[X]| \geq t] \leq \frac{\text{Var}[X]}{t^2}$ or equivalently, $\Pr[|X - \mathbb{E}[X]| \geq t \cdot \sigma] \leq \frac{1}{t^2}$ $(\sigma := \sqrt{\text{Var}[X]})$

Proof: left out of the sheet, it uses Markov on $Y := (X - \mathbb{E}[X])^2$

Diagram:



EXAMple from FS17: Let $n \in \mathbb{N}$, and let X be a random var. with $\mathbb{E}[X] = n \log n$ and $\text{Var}[X] = n^2$. Show: $\exists C > 0$ s.t. $\Pr[X \geq n \log n + C \cdot n] \leq 0.01$.

By Chebyshev, $\Pr[T_n \geq n \log n + C \cdot n] \leq \Pr[|T_n - n \log n| \geq C \cdot n]$

$= \Pr[|T_n - \mathbb{E}[T_n]| \geq C \cdot n]$

$\leq \frac{\text{Var}[T_n]}{(C \cdot n)^2} = \frac{n^2}{C^2 n^2} = \frac{1}{C^2}$.

You can choose $C \geq 10$, so $\Pr[T_n \geq n \log n + C \cdot n] \leq 1/C^2 \leq 0.01$.

Chernoff bounds:

(W9 - W10 starts here)

→ for a random var. X with $X \sim \text{Bin}(n, p)$, these are much more accurate than Markov or Chebyshev

Theorem: Let $X_1, \dots, X_n$ be independent Bernoulli-distributed random variables with $\Pr[X_i = 1] = p_i$ and $\Pr[X_i = 0] = 1 - p_i$. Then, for $X = \sum_{i=1}^{n} X_i$:

i) $\Pr[X \geq (1-\delta)\mathbb{E}[X]] \leq e^{-\frac{1}{2}\delta^2 \mathbb{E}[X]}$ for $0 < \delta \leq 1$

ii) $\Pr[X \leq (1-\delta)\mathbb{E}[X]] \leq e^{-\frac{1}{2}\delta^2 \mathbb{E}[X]}$ for $0 < \delta \leq 1$

iii) $\Pr[X \geq t] \leq 2^{-t}$ $\forall t \geq 2e\mathbb{E}[X]$

Coupon Collector: this is an important example to understand, similar algorithms may be featured on the exam

→ Consider n coupons, we receive one of them every round at random

→ Define the random variable $X :=$ amount of rounds until we have all n coupons

$\mathbb{E}[X] = ?$. → Consider n phases, where phase i is the round in which we own $i-1$ coupons, including the round in which we obtain the i-th coupon. $X_i :=$ amount of rounds in phase i

→ Then: $X = \sum_{i=1}^{n} X_i$, $X_i \sim \text{Geo}\left(\frac{n-(i-1)}{n}\right)$ ← $\Pr[\text{"round gives } i\text{-th coupon"}] = \frac{n-(i-1)}{n}$ ← new coupons / ← all coupons

↑ waiting for i-th success

→ $\mathbb{E}[X] \overset{\text{Lin. of } \mathbb{E}}{=} \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} \frac{n}{n-(i-1)} = n \sum_{i=1}^{n} \frac{1}{i} = n \cdot H_n = O(n \log n)$

($\mathbb{E}[Y] = \frac{1}{p}$ if $Y \sim \text{Geo}(p)$)

H_n: n-th harmonic number; $H_n := \sum_{i=1}^{n} \frac{1}{i} = \ln(n) + O(1)$

Bounding probabilities using **Markov**: We know $\mathbb{E}[X] = n \cdot H_n \leq n \cdot \ln n + n \leq 2 \cdot n \cdot \ln(n)$ (for $n \geq 3$)

Markov: $\Pr[X \geq 100 \cdot \ln(n)] \leq \frac{\mathbb{E}[X]}{100 \cdot \ln(n)} = \frac{2n \cdot \ln(n)}{100 \cdot \ln(n)} = \frac{n}{50}$ (prof's example on lecture notes was wrong)

Determining Var[X]:

→ $X = \sum_{i=1}^{n} X_i$, X_i = rounds in which we own $i-1$ unique coupons, $X_i \sim Geo(\frac{n-(i-1)}{n}) = Geo(1-\frac{i-1}{n})$,

$X_1, \dots, X_n$ independent

→ $Var[X_i] = \frac{1-p_i}{p_i^2} \leq \frac{1}{p_i^2} = \left(\frac{n}{n-i+1}\right)^2$

→ $Var[X] = Var[\sum_{i=1}^{n} X_i] \underset{\text{ind. of } X_1,\dots,X_n}{=} \sum_{i=1}^{n} Var[X_i] \leq \sum_{i=1}^{n} \left(\frac{n}{n-i+1}\right)^2 = n^2 \sum_{i=1}^{n} \frac{1}{i^2} \leq n^2 \cdot \frac{\pi^2}{6}$ ($\sum_{i=1}^{\infty} \frac{1}{i^2} = \frac{\pi^2}{6}$)

→ Example of **Chebyshev** for $t = n\sqrt{\ln n}$:

$\Rightarrow \Pr[|X - \mathbb{E}[X]| \geq n \cdot \sqrt{\ln n}] \leq \frac{\pi^2}{6 \ln(n)} \xrightarrow{n \to \infty} 0$

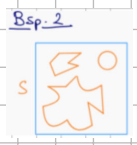
Target shooting algorithm (Chernoff 🎯):

→ given: Sets S, U with $S \subseteq U$, goal: Find $\frac{|S|}{|U|}$

→ $O(n)$ is too slow, we want a probabilistic outcome

→ Define ind. vars. $I_S : U \to \{0,1\}$ with $I_S(u) = 1 \Rightarrow u \in S$

→ Example applications: 1. U = a population, S = infected people, 2.



$U = [0,1] \times [0,1]$, what area does S have?

Idea: Test if N elements are in S

Algorithm: ① Select $u_1, \dots, u_N$ from U randomly and independently with equal probability distributions

② Return $\frac{1}{N} \sum_{i=1}^{N} I_S(u)$

Intuitively: bigger subset chosen ⇒ better approximation

Let ind. vars. $Y_i = I_S(u)$ for $i = 1, \dots, N$ and Y the algo.'s result $= \frac{1}{N} \cdot \sum_{i=1}^{N} Y_i$

→ $\mathbb{E}[Y] = \frac{1}{N} \sum_{i=1}^{N} \underbrace{\mathbb{E}[Y_i]}_{=\frac{|S|}{|U|}} = \frac{|S|}{|U|}$

Goal: $|Y - \frac{|S|}{|U|}| = |Y - \mathbb{E}[Y]|$ ← Error, we want it to be small with a high prob.

→ more precisely: for $\delta, \varepsilon > 0$: $\Pr[|Y - \frac{|S|}{|U|}| \leq \varepsilon \frac{|S|}{|U|}] \geq 1 - \delta$ (multiplicative error of $\leq (1 \pm \delta)$)

Theorem: Let $\delta, \varepsilon > 0$. If $N \geq 3 \frac{|U|}{|S|} \cdot \frac{1}{\varepsilon^2} \cdot \ln(\frac{2}{\delta})$, the returned variable Y of the target shooting algorithm is in the interval $[(1-\varepsilon)\frac{|S|}{|U|}, (1-\varepsilon)\frac{|S|}{|U|}]$ with probability $\geq 1 - \delta$.

Proof: $\Pr[|Y - \frac{|S|}{|U|}| > \varepsilon \cdot \frac{|S|}{|U|}] = \Pr[|Y - \mathbb{E}[Y]| \geq \varepsilon \cdot \mathbb{E}[Y]] = \Pr[|NY - \mathbb{E}[NY]| \geq \varepsilon \cdot \mathbb{E}[NY]]$

for which N does $\Pr[|NY - \mathbb{E}[NY]| \geq \varepsilon \cdot \mathbb{E}[NY]]$ hold.

→ $NY = Y_1 + \dots + Y_N$ = a sum of N independent Bernoulli variables → **Chernoff:**

→ $\Pr[|NY - \mathbb{E}[NY]| > \varepsilon \cdot \mathbb{E}[NY]] \leq 2 \cdot e^{-\frac{1}{3}\varepsilon^2 \cdot \mathbb{E}[NY]} = 2e^{-\frac{1}{3}\varepsilon^2 N \frac{|S|}{|U|}} \leq \delta$ (for $N \geq 3 \frac{|U|}{|S|} \cdot \frac{1}{\varepsilon^2} \cdot \ln(\frac{2}{\delta})$)

Randomized Algorithms:

How deterministic algorithms work:

Input $I \Rightarrow$ Alg. $\mathcal{A} \rightarrow$ Output $\mathcal{A}(I)$

$\rightarrow \mathcal{A}(I)$ is always correct and

$\mathcal{A}$ always has the same asymptotic runtime

How probabilistic algorithms work:

Input $I \Rightarrow$ Alg. $\mathcal{A} \Rightarrow$ Output $\mathcal{A}(I,R)$

$\uparrow R$

Random variable

$\rightarrow$ The output depends on I and $R \rightarrow$ runtime or correctness can be variable and are not reproducible

$\rightarrow$ In computational theory and ar analysis, we assume **physical random generation** of the random bits used to create truly random numbers (Exercise: count how many times I said "random")

$\rightarrow$ In practice, we use pseudo-random generators (deterministic) using a **seed**

$\rightarrow$ Example: MD5 hashsums

"Üppige Auswahl" routine:

$\rightarrow$ given $a_1, \dots, a_n \in \{0,1\}$, of which half equal 1, find an index i with $a_i = 1$

Trivial (and best) deterministic algorithm: for $i = 1, \dots, n$: if $a_i = 1$ return 1

Theorem: every deterministic alg. needs at least runtime $\Omega(n)$.

Proof (I love this argumentation style!): For any given algorithm and n we create an instance:

$\rightarrow$ What index does the algorithm choose? Place a 0 there ...

$\rightarrow$ repeats until $n/2+1$, which places the alg. in $\Theta(n)$.

Las Vegas Algorithm (don't worry, we will discuss this):

repeat:

  $i := \text{Uniform}(\{1, \dots, n\})$

  if $a_i = 1$; return i

$\Pr[$"in the first k repetitions, we have an index i with $a_i = 0$"$]$

$= \Pr[T > k]$ (T is the random var. for the amount of repetitions)

$= (1/2)^k$

Direct approach:

$\mathbb{E}[T] = \sum_{n=1}^{\infty} \Pr[T \geq n] = \sum_{n=1}^{\infty} \Pr[T > n-1] = \sum_{n=1}^{\infty} (1/2)^{n-1}$

$= \frac{1}{2} + \frac{1}{4} + \dots = 2$

Indirect approach: $\mathbb{E}[T] = \Pr[a_i = 1] \cdot 1 + \Pr[a_i = 0] \cdot (1 + \mathbb{E}[T'])$ (T' → amount of repetitions after first one fails) $\Rightarrow \mathbb{E}[T] = 1 + \mathbb{E}[T]/2 \Rightarrow \mathbb{E}[T] = 2$

$\rightarrow$ 2 repetitions: Algo runs in $O(1)$!

Monte Carlo and Las Vegas algorithms:

Monte Carlo algorithm: Correctness is a random variable, runtime is guaranteed

Las Vegas algorithm: Runtime is a random variable, Correctness is guaranteed

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→ Alternatively, we can see them as algos. that sometimes return '!???!' instead of a result (if they don't finish in time)

Reducing error:

Las Vegas: Goal: Given the Las Vegas algorithm A with $Pr[A(I) \text{ correct}] \geq \varepsilon$, find a Las Vegas algorithm A_δ with $Pr[A_\delta(I) \text{ correct}] \geq 1-\delta$.

Idea: repeat A a sufficient amount of time → how often?

Theorem: Let A be a randomized algorithm that never returns a wrong answer, but '!???!' at times with $Pr[A(I) \text{ correct}] \geq \varepsilon$. Then, for all $\delta > 0$: Let A_δ be the algorithm that runs A until a value other than '!???!' is output, in which case this value is returned, or until '!???!' has been output $N = \lceil \varepsilon^{-1} \ln \delta^{-1} \rceil$ times, in which case '!???!' is returned.

→ Reducing the probability of error only takes a constant amount of iterations → Example: $\varepsilon = 0.25$

δ	$N = \lceil \varepsilon^{-1} \ln \delta^{-1} \rceil$
0.1	10
0.01	19
0.001	28
0.0001	37
0.00001	47
0.000001	56

Monte Carlo: In general, it is not possible to reduce the probability of error

→ imagine a coinflip algorithm, you can't get a reliable answer

→ we can reduce the error prob. when there is a one-sided error or error prob. $< \frac{1}{2}$

→ one-sided error ($\circledast$): $Pr[A(I) = \text{YES}] = 1$ if I should return YES, $Pr[A(I) = \text{NO}] \geq \varepsilon$ if I should return NO

Theorem: Let A be a randomized algorithm that always returns YES or NO, following $\circledast$.
Then for all $\delta > 0$: Let A_δ be the algorithm that calls A until either NO is output and subsequently returned or YES is output $N = \lceil \varepsilon^{-1} \ln \delta^{-1} \rceil$ times and subsequently returned.

Theorem: Let $\varepsilon > 0$ and A be a randomized algorithm that always returns YES or NO with $Pr[A(I) \text{ correct}] \geq \frac{1}{2} + \varepsilon$. Then, for all $\delta > 0$: Let A_δ be the algorithm that calls A $N = \lceil 4\varepsilon^{-2} \ln \delta^{-1} \rceil$ times and returns the majority of received answers. Then $Pr[A_\delta(I) \text{ correct}] \geq 1 - \delta$.

Proof of Las Vegas theorem: $N = \lceil \frac{1}{\varepsilon} \cdot \ln \frac{1}{\delta} \rceil$. $Pr[\text{"N times '!???!'"}] \leq (1-\varepsilon)^N \underset{\uparrow 1+x \leq e^x}{\leq} e^{-\varepsilon N} \leq e^{-\varepsilon \cdot \frac{1}{\varepsilon} \cdot \ln \frac{1}{\delta}} = e^{\ln \delta} = \delta$.

Proof of second MC theorem: we know $0 < \varepsilon < \frac{1}{2}$ and define X_i: $\begin{cases} 1 & \text{if } i\text{-th call succeeds} \\ 0 & \text{else} \end{cases}$

$\left[X = \sum_{i=1}^{N} X_i \Rightarrow \mathbb{E}[X_i] = p \right]$ $\quad p = Pr[X_i = 1] = Pr[A(I) \text{ corr.}] \geq \frac{1}{2} + \varepsilon \Rightarrow \mathbb{E}[X] \overset{\text{(LoE)}}{=} N \cdot p = (\frac{1}{2} + \varepsilon) N$

Also: $\frac{N}{2} \underset{\uparrow 0 < \varepsilon < \frac{1}{2}}{\leq} (1-\varepsilon)(\frac{1}{2}+\varepsilon)N$. Now, compute $Pr[A_\delta(I) \text{ correct}]$:

$Pr[A_\delta(I) \text{ corr.}] \geq Pr[X > \frac{N}{2}] \underset{\uparrow \text{we will use Chernoff}}{=} 1 - Pr[X \leq \frac{N}{2}]$ $\quad || \quad Pr[X \leq \frac{N}{2}] \leq Pr[X \leq (1-\varepsilon)\mathbb{E}[X]] \underset{\uparrow \mathbb{E}[X] \geq \frac{N}{2} \geq \frac{1}{\varepsilon^2} \ln \frac{1}{\delta}}{\leq} e^{-\frac{1}{2}\varepsilon^2 \mathbb{E}[X]} \leq e^{-\ln \frac{1}{\delta}} = e^{\ln \delta} = \delta$

$\uparrow N = \lceil 4 \frac{1}{\varepsilon^2} \ln \frac{1}{\delta} \rceil$

Randomized algorithms for optimization problems:

Theorem: Let $\varepsilon > 0$ and A be a randomized algorithm for an optimization problem with $Pr[A(I) \geq f(I)] \geq \varepsilon$.

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Then, for all $\varepsilon > 0$: Let A_δ be the algorithm that calls A $N = \lceil \varepsilon^{-1} \ln \delta^{-1} \rceil$ times and returns the best of the outputted answers, for which $Pr[A_\delta \geq f(I)] \geq 1-\delta$

Prime Number Tests:

Problem: Given $n \in \mathbb{N}$, decide if n is prime $\Rightarrow$ if it has no divider in $\{2, \dots, n-1\}$. The usual number size is 1000 bits, the prime number theorem allows for a high success rate ($\approx \frac{1}{1000 \ln 2} \approx \frac{1}{693}$)

Prime Number Theorem: $\pi(x) := |\{n \in \mathbb{N} \mid n \leq x, n \text{ prime}\}| \sim \frac{x}{\ln x}$

Naive Algorithm: for all $a \leq \sqrt{n}$, test if a divides $n \rightarrow O(\sqrt{n})$, but we want $O(\log n)$

Note: I will lightly touch up on group theory for the following algorithms, for a deeper review, check out the DM sheet

Euclidean Test: let $\gcd(n,m)$ be the GCD of $n, m \in \mathbb{Z}$ which can be computed using Euclid's Algorithm in $O((\log_2 m)^3)$. Trivially, $\gcd(a,n) > 1$ for $a \in [n-1] \Rightarrow n$ not prime

Euclidean Prime Test(n):

1. Choose $a \in [n-1]$ uniformly and randomly
2. if $\gcd(a,n) = 1$ return 'prime'
3. else return 'not prime'

→ if n prime → always correct

→ if n not prime → wrong answer 'prime' with prob. $\frac{|\mathbb{Z}_n^*|}{n-1}$

→ $\mathbb{Z}_n^* = \{a \in [n-1] \mid \gcd(a,n) = 1\}$, $\varphi(n) := |\mathbb{Z}_n^*|$ ↑ Euler's Totient

→ n prime: $\mathbb{Z}_n^* = [n-1]$, $\varphi(n) = n-1$, $n = p^2$, p prime: $\varphi(n) = p(p-1) = n - \sqrt{n}$ ($\Rightarrow \frac{|\mathbb{Z}_n^*|}{n-1} \approx 1 - \frac{1}{\sqrt{n}}$)

→ $\mathbb{Z}_n^*$ with multiplication generates a group of order $\varphi(n)$

→ **Lagrange's Theorem:** H subgroup of $G \Rightarrow |H|$ divides $|G|$

→ $\text{ord}(a) := \min\{i \in \mathbb{N} \mid a^i = 1\}$ → order of subgroup $\langle a \rangle$ generated by $a := \{a^0 = 1, a^1, a^2, \dots\}$

→ $\text{ord}(a)$ divides $|G|$. Therefore, $a^{|G|} = a^{\text{ord}(a)\frac{|G|}{\text{ord}(a)}} = 1^{\frac{|G|}{\text{ord}(a)}} = 1$

→ in $\mathbb{Z}_n^*$: $a^{\varphi(n)} = 1$ for all $a \in \mathbb{Z}_n^*$ ($\varphi(n) = |\mathbb{Z}_n^*|$)

→ in $\mathbb{Z}_n^*$, n prime: $a^{n-1} = 1$ for all $a \in \mathbb{Z}_n^*$ ($\varphi(n) = n-1$)

→ **Fermat's Little Theorem:** if $n \in \mathbb{N}$ prime, for all numbers $a \in [n-1]$: $a^{n-1} \equiv_n 1$ ($a^{n-1} = 1$ in $\mathbb{Z}_n^*$)

Fermat Prime Number Test(n):

1. Uniformly and randomly choose $a \in [n-1]$
2. if $a^{n-1} \equiv_n 1$ return 'prime' → $a^{n-1} \equiv_n$ can be sped up with binary exponentiation (BigInteger.modPow is fast)
3. else return 'not prime'

→ n prime: always correct, n not prime: wrong answer 'prime' with prob. $\leq \frac{1}{2}$ unless n is a Carmichael number

→ repeat k times if 'prime' is output → bring error down to $\leq \frac{1}{2^k}$ → Error: n not prime and $a^{n-1} \equiv_n 1$ ↑ pseudo-prime base

$PB_n := \{a \in [n-1] \mid a^{n-1} \equiv_n 1\} = \{a \in \mathbb{Z}_n^* \mid a^{n-1} \equiv_n 1\}$ (pseudo-prime bases)

→ n is a Carmichael number if it is not prime and $PB_n = \mathbb{Z}_n^*$. Examples: $561 = 3 \cdot 11 \cdot 17$, $1105 = 5 \cdot 13 \cdot 17$

→ PB_n subgroup of $\mathbb{Z}_n^*$ because $a^{n-1} \equiv_n 1 \wedge b^{n-1} \equiv_n 1 \Rightarrow (ab)^{n-1} \equiv_n 1$

→ if $PB_n \neq \mathbb{Z}_n^*$ (n not Carmichael), $|PB_n|$ divides $|\mathbb{Z}_n^*|$ nontrivially and $|PB_n| \leq \frac{\varphi(n)}{2} \leq \frac{n-1}{2}$

Miller-Rabin Certificate:

→ for n prime, $(\mathbb{Z}_n, \oplus, \odot, 0, 1)$ is a field, meaning that the equation $x^2 = 1$ has two solutions $\Rightarrow 1$ and $-1 (= n-1)$

→ Let $n > 2$ odd. Write $n - 1 = 2^k d$ with $k, d \in \mathbb{N}$, d odd. If n prime and $a \in [n-1]$:

→ $a^{2^k d} = a^{n-1} \equiv_n 1 \overset{\text{(Fermat)}}{\Rightarrow} a^{2^{k-1} d} \equiv_n 1$ or $n-1 \overset{\text{(def. field)}}{\Rightarrow}$ if $a^{2^{k-1} d} \equiv_n 1$, $a^{2^{k-2} d} \equiv_n 1$ or $n-1$

$\Rightarrow$ if $a^{2d} \equiv_n 1$, $a^d \equiv_n 1$ or $n-1 \Rightarrow a^d \equiv_n 1$ or $\exists i \in \{0, \dots, k-1\} : a^{2^i d} \equiv_n n-1$.

→ This is the certificate → Example: 561 (Carmichael): $2^{560} \equiv_{561} 1$, $2^{280} \equiv_{561} 1$, $2^{140} \equiv_{561} 67$

$\Rightarrow 2$ is a M-R certificate showing that 561 is not prime

→ $a \in \{2, 3\}$ is a M-R certificate for all $n < 1\,373\,653$

Algorithm (Miller-Rabin prime test(n) for $n > 1$, odd):

1. Express $n - 1 = 2^k d$ with $d, k \in \mathbb{N}$, d odd
2. Uniformly, randomly choose $a \in [n-1]$
3. if $a^d \equiv_n 1$ or $\exists i \in \{0, \dots, k-1\} : a^{2^i d} \equiv_n n-1$ then return 'prime'
4. else return 'not prime'

→ n prime: always correct, n not prime: wrong answer 'prime' with prob. $\leq \frac{1}{2}$ (in reality $\leq \frac{1}{4}$)

→ I am purposefully omitting the proofs.

QuickSort:

```
Quicksort(A,l,r)
if l<r then:
    p ← Uniform({l, l+1, ..., r})   // randomize pivot
    t ← Partition(A,l,r,p)          // "sort" A[l],...,A[r] s.t. the elements left of A[p] are < A[p] and the elements right of A[p] are > A[p] ⇒ t is the new position of A[p]
    QuickSort(A,l,t-1)
    QuickSort(A,t+1,r)
```

Runtime analysis (exam relevant): Runtime → $O(T_{1,n})$ with $T_{l,r}$ being a random variable counting the amount of comparisons made during Quicksort(A,l,r)

Theorem: $\mathbb{E}[T_{1,n}] \leq 2(n-1)\ln(n) + O(n)$ ← constant

Proof: for $l \geq r$: $T_{l,r} = 0$, for $l < r$: $\mathbb{E}[T_{l,r}] = \sum_{i=l}^{r} P[t=i] \cdot \mathbb{E}[T_{l,r} \mid t=i] = \sum_{i=l}^{r} \frac{1}{r-l+1} \cdot (r - l + \mathbb{E}[T_{l,i-1}] + \mathbb{E}[T_{i+1,r}])$

(Uniform prob.; Partition makes $r-l$ comparisons; recursive calls; elements to be sorted)

Observation: $\mathbb{E}[T_{l,r}]$ depends on the integer $r-l+1$

For $t_n = \begin{cases} 0 & \text{if } n \leq 1 \\ \frac{1}{n}\sum_{i=0}^{n-1}(n-1 + t_i + t_{n-i-1}) & \text{if } n \geq 2 \end{cases}$, $\mathbb{E}[T_{l,r}] = t_{r-l+1}$ (proven by induction)

Goal: an upper bound for $\mathbb{E}[T_{1,\dots,n}] = t_n \rightarrow \forall n \geq 3: n \cdot t_n = \sum_{i=0}^{n-1} (n-1 + t_i + t_{n-i-1})$ (1)

$$(n-1) \cdot t_{n-1} = \sum_{i=0}^{n-2} (n-2 + t_i + t_{n-i-2}) \quad (2)$$

$$(1)-(2): n \cdot t_n - (n-1) t_{n-1} = 2(n-1) + 2 \cdot t_{n-1}$$

$$\Rightarrow t_n = \frac{n+1}{n} \cdot t_{n-1} + \frac{2(n-1)}{n} \leq \frac{n+1}{n} t_{n-1} + 2 \quad (*)$$

Proposition: $\forall n \geq 2: t_n \leq 2 \sum_{i=3}^{n+1} \frac{n+1}{i}$

Proof by induction over n: Base Case: $n=2 \rightarrow t_2 = 1 \leq 2 = 2 \sum_{i=3}^{3} \frac{3}{i}$

$$n \geq 3: 2 \sum_{i=3}^{n+1} \frac{n+1}{i} = 2 \sum_{i=1}^{n} \frac{n+1}{i} + 2 \frac{n+1}{n+1} = \underbrace{2 \sum_{i=3}^{n} \frac{n}{i}}_{\geq t_{n-1}} + 2 \overset{(*)}{\geq} t_n \quad \blacksquare$$

QuickSelect: given a sequence (array) $A[1], \dots, A[n]$ with pairwise distinct values, find the k-th smallest value

QuickSelect(A, l, r, k):

oh yeah, W11-W13 begins here

```
p ← Uniform(l, l+1, .., r)
t ← Partition(A, l, r, p) // p is the t-th largest element
if t = l+k-1 then:
    return A[t]
else if t > l+k-1 then:
    return QuickSelect(A, l, t-1, k) // target is to the left of t
else
    return QuickSort(A, t+1, r, k) // target is to the right of t
```

Expected runtime: $O(T)$, T is the random variable tracking the amount of comparisons (also exam relevant!)

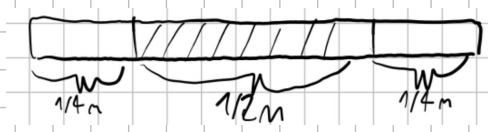
→ QuickSelect(A, 1, n, k) causes N (random variable) calls of Partition(A, l_i, r_i, p) for a random sequence $(l_0, r_0, k_0), \dots (l_N, r_N, k_N)$ with $(l_0, r_0, k_0) = (1, n, k)$. Therefore, $(l_{i+1}, r_{i+1}) = (l_i, t-1)$ or $(t+1, r_i)$, where t is a uniformly distributed random variable

→ Runtime of QuickSelect(A, 1, n, k) → $T = \sum_{i=1}^{N} (r_i - l_i)$ ← Partition(A, l_i, r_i, p) takes $r_i - l_i$ comparisons

→ We define N_j as amount of QuickSelect calls, for which $(3/4)^j n < r_i - l_i + 1 \leq (3/4)^{j-1} n$.

Then, $T \leq \sum_{j=1}^{\infty} N_j \cdot \underbrace{(3/4)^{j-1} n}_{\text{upper bound for comparisons of n partition calls}} \overset{\text{LoE}}{\Rightarrow} \mathbb{E}[T] \leq n \cdot \sum_{j=1}^{\infty} \mathbb{E}[N_j] \cdot (3/4)^{j-1}$.

Observation: If the pivot lies in the middle $\frac{r-l+1}{2}$ elements, the amount of considered ($r_i - l_i + 1$) elements reduces to at least a factor of 3/4



$\Rightarrow$ this has a probability of 1/2 and is exactly what N_j measures $\Rightarrow \mathbb{E}[N_j] \leq 2$ for all $j \geq 1$

$$\Rightarrow \mathbb{E}[T] \leq 2n \sum_{j=1}^{\infty} (3/4)^{j-1} = 2n \sum_{j=0}^{\infty} (3/4)^j = 2n \cdot 4 = 8n$$

$\Rightarrow$ QuickSelect runs in linear time, which was our goal

Duplicates and HashMaps:

Problem: A dataset $D = (s_1, \dots, s_n)$ is a sequence of n elements. A tuple (i, j) with $1 \leq i < j \leq n$ is a duplicate

in D if $s_i = s_j$. Given a dataset D, find all duplicates.

Example: $D = (A, C, B, Z, C, B, C)$, $\underset{1}{}\ \underset{2}{}\ \underset{3}{}\ \underset{4}{}\ \underset{5}{}\ \underset{6}{}\ \underset{7}{}$, set of duplicates in D: $Dupl(D) = \{(2,5), (2,7), (5,7), (3,6)\}$

Simpler algorithm: Sort $((s_1,1), \dots, (s_n,n))$ by their first coordinate, iterate over the sorted sequence to find duplicates

Example:

Indexing:	(A,1)	(C,2)	(B,3)	(Z,4)	(C,5)	(B,6)	(C,7)
Sorting:	(A,1)	(B,3)	(B,6)	(C,2)	(C,5)	(C,7)	(Z,4)
Duplicates:			(3,6)		(2,5), (2,7), (5,7)		

Analysis: Sorting takes $O(n \log n)$ comparisons and memory access, iterating takes $O(n + |Dupl(D)|)$

→ **Reason for Hashmaps:** It is more efficient to compare hash values than large objects

→ Let the elements in D be from a universe U. We use a hash function $h: U \to [m]$ and require:

→ h is efficiently computable

→ h acts as a randomized function, meaning $\forall u \in U\ \forall i \in [m]: \Pr[h(u) = i] = \frac{1}{m}$

→ Important: Each $h(s_i)$ is randomized over $[m]$, but $s_i = s_j \Rightarrow h(s_i) = h(s_j)$, we compress by choosing m much smaller than U

Example solution:

Input:	A	C	B	Z	C	B	C
Hash and index:	(31,1) h(A)	(27,2) h(C)	(12,3) h(B)	(12,4) h(Z)	(27,5) h(C)	(12,6) h(B)	(27,7) h(C)
Sort:	(12,3)	(12,4)	(12,6)	(27,2)	(27,5)	(27,7)	(31,1)
Duplicates:		(3,6),		(2,5), (2,7), (5,7)			

→ Important: check for collisions while finding duplicates, such as $h(B) = h(Z)$

→ Collisions are the new, unwanted duplicates in a hashmap $\Rightarrow$ pairs (i,j), $1 \le i < j \le n$ with $s_i \neq s_j$ and $h(s_i) = h(s_j)$

Probability of a collision: Let K_{ij} be the indic. var. of „(i,j) is a collision". Then:

$\Pr[K_{ij} = 1] = \begin{cases} \frac{1}{m} & \text{if } s_i \neq s_j \\ 0 & \text{if } s_i = s_j \end{cases} \Rightarrow \mathbb{E}[K_{ij}] \le \frac{1}{m}\ \forall i,j \Rightarrow \mathbb{E}[\#\text{collisions}] \le \sum_{1 \le i < j \le n} \mathbb{E}[K_{ij}] \le \binom{n}{2} \cdot \frac{1}{m}$ ($\le \frac{1}{m}$; pairs with $i<j$ are half of total)

$\Rightarrow$ for $m = n^2$, $\mathbb{E}[\#\text{collisions}] < 1$, which only adds a constant factor to the runtime

Runtime: n hash function computations, $O(n \log n)$ to sort $\lceil 2 \log n \rceil$-bit integers, $|Dupl(D)| + \#\text{collisions}$

$\Rightarrow$ same asymptotic runtime as simple algorithm but the comparisons themselves are less expensive for more complex objects

General Algorithms: (W10-W13 starts here)

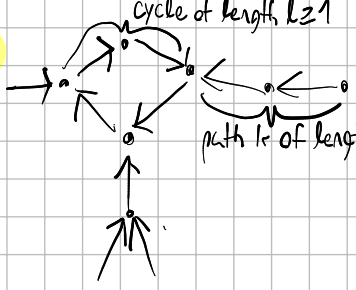
Floyd's cycle finding alg. (Hare & Tortoise):

Given an array $a[1, \dots, n]$ with $a[i] \in \{1, \dots, n-1\}$ ($n-1$ unique values), find indices $i \neq j$ with $a[i] = a[j]$ in $O(n)$ time, $O(1)$ space: doing one of each is easy: optimize space by remembering previous values in $O(n^2)$ time or use BucketSort for $O(n)$ time and space.

→ Consider the directed graph $D = (V, A)$ with $V = [n]$, $A = \{(i, a[i]) \mid 1 \le i \le n\}$

→ how does D look? each vertex has one outgoing and one ingoing edge → the subgraph D_n consists of vertices reachable from n

→ **Example:**



→ consider the sequence of vertices $x_0 = n$, $x_t := a[x_{t-1}]\ \forall t \ge 1$

→ Claim: $\exists t \le n$ s.t. $x_t = x_{2t}$

→ Proof (optional): Let $r = \lceil \frac{t}{k} \rceil \cdot k - t \ge 0$. Then x_{t+r} is in the cycle C.

$\Rightarrow t + r = \lceil \frac{t}{k} \rceil \cdot k$ is an integer multiple of k (length of C) $\Rightarrow x_{t+r} = x_{2(t+r)}$ ✓